

Physics-augmented neural networks and projection-based reduced-order models for anisotropic hyperelasticity

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Abstract

Physics-augmented neural networks (PANNs) for hyperelastic constitutive modelling are commonly held to a six-point admissibility checklist: thermodynamic consistency, objectivity, tensor symmetry, a stress-free reference state, polyconvexity, and a growth condition [1, 2]. Existing constructions achieve all six only for a material with a named symmetry group, or give up one property to handle general anisotropy; whether this trade-off is fundamental is an explicit open problem in the recent literature. This paper resolves it for a two-dimensional anisotropic representative volume element (RVE): an energy-based PANN, built from an input-convex neural network (ICNN) acting on balanced directional invariants of $\mathbf{F}$ and $\text{cof } \mathbf{F}$, satisfies all six properties exactly and by construction, with closed-form proofs rather than training penalties, including an explicit polyconvexity construction in the sense of Ball [2]. The same proofs transfer verbatim to a second convex core, an input-convex Kolmogorov–Arnold network (ICKAN). Benchmarked on a test trajectory against a projection-based hyper-reduced-order model in both its Newton-corrected (HPROM–ANN) and direct (D-HPROM–ANN) operating modes, and against a direct regression baseline, the admissibility certificate proves necessary in practice, because the uncertified regression baseline matches or exceeds every certified model’s test-trajectory accuracy while directly violating thermodynamic consistency, and a matched uncertified flexible PANN fails a rank-one convexity check on 7% of sampled deformation states despite equally strong accuracy. Deployed as the macroscopic material law of an unrelated structural benchmark, Cook’s membrane, this same distinction determines whether the structural solve itself succeeds. The certified models converge in three Newton iterations at every ramped load step, while the uncertified flexible PANN fails to converge at the very first step – a real structural consequence of the missing certificate, even though the identical rank-one check finds no violation at the specific states this particular problem visits. Nesting the same reduced-order surrogates, and, once, the true, non-reduced RVE itself, as the actual microscale solver at every Gauss point – a genuine FE^2 computation, not a direct deployment – confirms this robustness result, and, by comparing every model directly against the true multiscale answer for the first time rather than only against each other, exposes a small-strain accuracy gap in the certified cores that the test-trajectory check misses; one additional, cheap loss term, matching each model’s own tangent stiffness at zero strain to the RVE’s true linear-elastic limit, corrects most of it.

1 Introduction

Multiscale (FE^2 -type) computational homogenization resolves each macroscopic material point by solving a nonlinear boundary value problem on a representative volume element (RVE), so that the macroscopic constitutive response is not postulated but computed from the microstructure directly. This is expensive. Every macroscopic Gauss point, at every load step and every Newton iteration, requires its own converged microscale solve. Two largely separate lines of work have grown up around reducing this cost, and they sit on opposite sides of the classic *non-intrusive/intrusive* divide in reduced-order and surrogate modelling.

The non-intrusive route treats the microscale solve as a black box: precompute enough RVE solutions offline, then learn a direct surrogate for the homogenized constitutive law itself, so that no microscale solve is needed online at all. Early versions of this idea learn a bare regression map from strain to stress; the

physics-augmented neural network (PANN) literature that has grown from it [1, 20, 3] instead builds physical structure into the learned energy potential itself: thermodynamic consistency, objectivity, tensor symmetry, a stress-free reference state, polyconvexity, and a growth condition, so that the surrogate is trustworthy by construction, not merely accurate where it happens to have been trained.

The intrusive route, by contrast, leverages prior knowledge of the RVE’s own governing equations to construct a structured low-dimensional representation of its state space, rather than treating the microscale solve as a black box. A projection-based reduced-order model (PROM) restricts the unknown displacement field to a low-dimensional subspace identified by proper orthogonal decomposition (POD) of precomputed snapshots, and projects the governing equilibrium equations onto that same subspace (Galerkin projection) to obtain a small, closed system of equations, tracing back to the reduced-model multiscale method of Yvonnet and He [6] and developed since for nonlinear multiscale structural problems [7, 8, 9, 10, 11]. A single linear subspace, however, generally cannot compress a genuinely nonlinear problem’s solution manifold efficiently. Its Kolmogorov n -width decays slowly, which restricts the efficiency of linear-subspace reduction for a problem this nonlinear, since achieving good accuracy can require keeping the retained dimension itself large. This cost does not stop at reconstruction accuracy, because hyperreduction’s own online cost scales with the reduced mesh size and the retained dimension together, so keeping that dimension large also keeps the reduced mesh of retained elements large and erodes the very speedup hyperreduction is meant to provide. More recent work addresses the first cost by correcting the subspace itself. Efforts here are diverse. Notable examples partition it into a library of local reduced bases selected online according to the current state [12], replace it with a single quadratic correction [13], or learn a more expressive nonlinear-manifold correction with a deep autoencoder, evaluated online through a manifold Galerkin or manifold least-squares Petrov–Galerkin projection [14, 15]. This paper instead builds on the PROM-ANN/HPROM-ANN framework of Barnett, Farhat and Maday [16] (later extended to interpretable Gaussian-process and radial-basis-function closures, dubbed PROM-GPR/RBF, by Ares De Parga et al. [38]), which models the correction as the closure error of the original affine subspace instead of replacing the subspace outright. The second cost is addressed by replacing the full Gauss-point residual assembly with a sparse, learned rule from the *project-then-approximate* hyperreduction family, such as the empirical cubature method (ECM) [18] and/or the energy-conserving sampling and weighting (ECSW) method [39, 40, 41]. This same PROM-ANN/RBF/GPR family has shown that pairing ECSW hyperreduction with a closure-based approximation manifold clearly shrinks the reduced mesh needed, against both the linear (affine) approximation and other nonlinear manifolds such as the quadratic one [13, 16, 17, 38]. This project’s own RVE, however, is solved once per macroscopic Gauss point rather than once overall, so even a reduced mesh already shrunk this way is repeated many times over online. Hernández asks whether a commensurate reduction can be pushed further still [19], and answers it with the manifold-adaptive-weight extension of ECM (MAW–ECM), which lets the cubature weights themselves vary along the low-dimensional manifold instead of staying fixed. Demonstrated there with a PROM-RBF decoder and ECM specifically, though the underlying reduced-mesh problem is the same one ECSW solves, MAW–ECM, used throughout this paper, compresses the already-shrunk reduced mesh by a further 80–97%. Throughout this route, the true microscale equilibrium equations are still being solved, just in compressed and hyperreduced form. The RVE’s own physics is never bypassed.

Both routes are active research areas in their own right, and both are used in this paper, not as competitors but as two independent, non-interchangeable sources of ground truth and comparison for the same RVE, evaluated on the same test load path: a rare case of the non-intrusive and intrusive philosophies being confronted with exactly the same problem rather than argued for in isolation.

For the non-intrusive, PANN route specifically, a recurring difficulty motivates the deeper focus of this paper. Physics-augmented neural constitutive models are commonly held to an admissibility checklist: (C1) thermodynamic consistency, (C2) objectivity, (C3) tensor symmetry, (C4) a stress-free reference state, (C5) polyconvexity, and (C6) a growth condition. Table 1 checks eight directly comparable constructions against this list, and no two broad strategies in the current literature yet achieve all six properties together for a general anisotropic material.

The first keeps polyconvexity exact but ties it to a named symmetry group. Klein et al. [1] and Linden et al. [20] give exact, architecturally-certified constructions for isotropic and transversely isotropic materials via a structural-tensor integrity basis, recently widened to cubic and tetragonal symmetry classes [24], but Linden et al. state explicitly that a complete, polyconvexity-compatible invariant set “may not exist for some symmetry groups . . . it might be possible that there does not exist a set of invariants to describe the anisotropy of materials with arbitrary microstructures,” and flag a general, symmetry-agnostic construction as an open problem. The second keeps a more flexible or general anisotropy representation but gives up something else. Kalina et al. [25] learn the structural tensor’s own parameters jointly with the network, more flexible than a fixed structural tensor, but explicitly decline polyconvexity, “too restrictive, especially for multiscale modeling”; As’ad, Avery and Farhat [3] instead substitute a strictly stronger condition, convexity of W in $\mathbf{E}$ (equivalent to a global SPD tangent), which Section 4.6.2 shows is measurably false for this RVE’s own ground

truth. A third cluster gets polyconvexity right, via the same additive, block-decoupled sufficient condition used here, but restricts anisotropy to at most two fixed fiber directions and rarely exercises the growth condition (C6) in practice, since the materials studied are (nearly) incompressible [21, 22, 23]. Section 4.6.4 onward discusses all eight works against every checklist item in the detail Table 1 only summarizes.

Whether this trade-off is fundamental, or an artifact of how the anisotropy has been encoded so far, is the central question this paper answers for a concrete, fully worked 2D RVE: a Neo-Hookean solid with a hole, admitting no assumed material symmetry.

Table 1: Comparison against the admissibility checklist. “Exact” means satisfied by construction for every parameter value, not approximated by training or data augmentation. “Group-free anisotropy” means the model represents a general anisotropic material without committing to a named symmetry class (isotropic, transversely isotropic, cubic, orthotropic, . . .) or a learned member of one. “Growth exercised” means the model’s volumetric term actually diverges/was actually tested at $J \rightarrow 0^+$, as opposed to being stated as a requirement but bypassed because the worked examples are (nearly) incompressible.

Work	Exact obj.	Exact ref. state	Ball polyconvex	Group-free anisotropy	Growth exercised
As’ad et al. [3]	yes	yes	no ($\mathbf{E}$ -convex)	yes (raw $\mathbf{E}$)	n/a
Linka & Kuhl [21]	yes	yes	yes	no (isotropic)	no (incompressible)
Tac et al. [22]	yes	partial ¹	yes	no (2 fixed fibers)	no (incompressible)
Klein et al. [1]	yes/approx. ²	approx. only	yes	no (named group)	yes
Thakolkaran et al. [23]	yes	yes	yes	no (isotropic)	no (no barrier term)
Linden et al. [20]	yes	yes	yes	no (named group)	yes
Klein et al. [24]	yes	yes	yes	no (named group) ³	n/a ⁴
Kalina et al. [25]	yes	yes	no	partial ⁵	yes
This work	yes	yes	yes	yes	yes

Answering that question carefully also requires knowing, precisely, what “accurate” does and does not certify. A model can match or exceed a certified model’s test-trajectory accuracy while silently violating thermodynamic consistency or losing rank-one convexity at ordinary deformation states; this is not a hypothetical concern but something demonstrated directly on this project’s own data in Sections 4.2.1 and 6.5. This is also why the paper evaluates six model variants side by side on the identical test path, spanning a genuine spectrum of embedded physical structure: a pure regression surrogate with no potential at all, a free (uncertified) energy surrogate, two independently certified polyconvex cores (ICNN and ICKAN), and the two operating modes of HPROM-ANN, a projection-based model that satisfies every admissibility property automatically because it solves (a hyperreduced form of) the true microscale equilibrium equations rather than learning a constitutive law at all.

Contributions. This paper’s own contributions are specifically:

1. A closed-form construction of a PANN for a general, group-free anisotropic hyperelastic material that satisfies (C1)–(C6) exactly, by construction, including exact polyconvexity in Ball’s sense (Section 4.6.4), to our knowledge the first construction to combine exact objectivity, an exact stress-free reference state, and exact polyconvexity without assuming or exploiting any named material symmetry class (Section 4.6.4). We also verify the correct necessary consequence of polyconvexity (rank-one convexity) numerically, and show that a strictly stronger, globally SPD-tangent condition used elsewhere in the literature for a different material is not required by polyconvexity and is, for this RVE’s own ground truth, false (Section 4.6.2).
2. A direct, quantitative demonstration that test-trajectory accuracy alone cannot certify admissibility, because a pure-regression baseline with no energy potential is the *most* accurate of all six model variants while directly violating (C1) (Section 4.2.1), and an otherwise-identical uncertified free model matches the certified models’ accuracy while failing a rank-one convexity check on 7% of sampled states (Section 6.5).

¹Exact for their CANN/NODE cores; an unaddressed gap for their ICNN core, whose biases are unconstrained.

²Exact for their invariant model W^I ; only approximated by data augmentation for their deformation-gradient model W^F .

³Their own Table 1 lists the trivial group C_1 ($|\mathcal{G}| = 1$) among the finite groups covered; substituting it collapses their group-symmetrization operator to a single, un-symmetrized term, illustrating how general their underlying integrity basis already is. Every one of their worked examples (Section 5), however, uses the cubic group specifically; Section 4.6.4 discusses this construction alongside our own, differently-scoped result for the general case.

⁴Not independently re-verified for this specific paper; not central to the group-free-anisotropy comparison drawn here.

⁵Anisotropy *parameters* (e.g. fiber directions) within a pre-chosen symmetry class are learned; the class itself is still user-specified.

3. Verification that the same certificate transfers, proof-for-proof, to a second convex core (an input-convex Kolmogorov–Arnold network, Section 6.4) at comparable accuracy with substantially fewer parameters.
4. A genuine FE² demonstration, in the same spirit as As’ad et al. [3] and Klein et al. [1], that the certificate’s value is not confined to the standalone constitutive surrogate (Section 6.6). Every model this paper builds, the projection-based ROM/HPROM surrogates of Section 5, and the true, non-reduced RVE FOM itself are all nested as the actual microscale solver at every macroscopic Gauss point of a large-deformation Cook’s-membrane benchmark, compared directly against the true multiscale answer rather than only against each other for the first time. The certified cores and both ROM/HPROM surrogates match the true FOM’s own Newton convergence behavior; the uncertified free model does not.

The full-order model, implemented in Kratos Multiphysics [5], the linear and nonlinear projection-based reduced models, and the manifold-adaptive-weight empirical-cubature hyperreduction used to build HPROM–ANN (Sections 2 and 5) are established tools from the literature, applied here but not contributed by this work; they are cited to their sources throughout.

Outline. Section 2 introduces the RVE and its FOM, fixing notation for both. Section 3 states the admissibility checklist precisely. Section 4 fixes kinematic and stress conventions and then takes each checklist item in turn, as Sections 4.2–4.7, with Section 4.6.4 giving the explicit polyconvex construction and, alongside it, the detailed comparison against the eight directly comparable PANN constructions of Table 1, establishing exactly what is already proven, what is imprecise, and what remains open in each. Section 5 introduces the PROM and HPROM–ANN pipeline used as an independent, non-ANN-constitutive-law comparison point, fixing notation for both. Section 6 describes the training loss and protocol, reports the test-trajectory comparison of all six model variants, repeats the polyconvexity proofs for a second, ICKAN core, and uses an uncertified baseline to show directly what the certificate protects against. Section 6.6 deploys every model this paper builds, plus the true, non-reduced RVE FOM itself, as the macroscopic material law of an unrelated structural benchmark, Cook’s membrane, as a genuine FE² demonstration: the ROM/HPROM surrogates of Section 5 and the true FOM are nested as the actual microscale solver at every Gauss point, not merely deployed directly like the four PANN tiers. Section 7 states the scope limits of this study explicitly, and Section 8 concludes.

2 Background: the representative volume element and its full-order model

This section fixes notation and gives a self-contained, high-level account of the RVE and its full-order model (FOM), which every model in this paper (the PANN as well as the projection-based and hyper-reduced comparison models of Section 5) is ultimately trained on, evaluated against, or built from.

2.1 The representative volume element and its full-order model

The RVE is a square unit cell with a central circular hole, Fig. 2, discretized by a standard displacement-based finite element mesh and governed by a single homogeneous, isotropic, compressible Neo-Hookean solid ($E = 1.628$ GPa, $\nu = 0.4$). A macroscopic strain $\boldsymbol{\varepsilon} = (E_{11}, E_{22}, \gamma_{12})$ is imposed through an affine Dirichlet condition on the outer boundary nodes (Fig. 2, red), while the remaining boundary and interior nodes, including the nodes on the hole, are left free. We write $\mathbf{u}$ for the full nodal displacement field over every node of the mesh (this is what is plotted in Fig. 5), partitioned into its Dirichlet and free parts as $\mathbf{u} = (\mathbf{u}_d, \mathbf{u}_f)$. On the Dirichlet boundary, $\mathbf{u}_d$ is prescribed exactly, node by node, by the affine macro-strain map determined by $\boldsymbol{\varepsilon}$. This map is affine in the reference-configuration position $\mathbf{X}$ of each Dirichlet node for a *fixed* $\boldsymbol{\varepsilon}$, but *not* a linear function of $\boldsymbol{\varepsilon}$ itself, since (made explicit below) it is built from the deformation gradient $\mathbf{F}(\boldsymbol{\varepsilon})$, a nonlinear (matrix-square-root) function of the three components $E_{11}, E_{22}, \gamma_{12}$, while $\mathbf{u}_f \in \mathbb{R}^{n_L}$, the free-node values, are unknown. The finite element (Galerkin) discretization of the governing balance of linear momentum reduces, in the quasi-static setting used here (no inertia, no explicit time dependence, since the test load path is a pseudo-time sequence of equilibrium states, not a genuine initial-value problem), to the nodal equilibrium residual

$$\mathbf{r}(\mathbf{u}_f; \boldsymbol{\varepsilon}) := \sum_{e=1}^{n_{el}} \mathbf{L}_e^T \left(\sum_{g=1}^{n_{gp}^{(e)}} \mathbf{B}_{e,g}^T \mathbf{S}_{e,g}(\mathbf{u}_f; \boldsymbol{\varepsilon}) w_{e,g} \right) = \mathbf{0}, \quad (1)$$

where $\mathbf{L}_e$ is the Boolean element-assembly operator, $\mathbf{B}_{e,g}$ is the element strain-displacement operator, $\mathbf{S}_{e,g}$ is the second Piola–Kirchhoff stress at Gauss point g of element e (evaluated from the true microscopic Neo-Hookean law, not a surrogate, and depending on $\mathbf{u}_f$ and the fixed, prescribed $\boldsymbol{\varepsilon}_d$ together), and $w_{e,g}$ is the corresponding FE quadrature weight (parent-domain weight times the isoparametric Jacobian determinant). Equation (1) is solved by Newton–Raphson for $\mathbf{u}_f$ at every prescribed $\boldsymbol{\varepsilon}$ along a load path; this is the FOM, and it is what generates every training snapshot and every test-trajectory ground-truth number (energy, stress, and their components) used elsewhere in this paper. Throughout this paper, $\boldsymbol{\varepsilon}$ denotes this prescribed, exact boundary strain. The deformation gradient $\mathbf{F} = \mathbf{U} = \sqrt{2\mathbf{E}(\boldsymbol{\varepsilon}) + \mathbf{I}}$ (the unique symmetric, rotation-free square root) is used to build the affine map $\mathbf{u}_d(\mathbf{X}) = (\mathbf{F} - \mathbf{I})\mathbf{X}$ directly, so the outer boundary’s own Green–Lagrange strain equals $\boldsymbol{\varepsilon}$ exactly, by construction, unlike the homogenized macro-stress, which is a computed output with more than one admissible definition. For $\mathbf{u}_d$ to be well-defined at all, the prescribed $\boldsymbol{\varepsilon}$ must satisfy $\mathbf{C}(\boldsymbol{\varepsilon}) := \mathbf{I} + 2\mathbf{E}(\boldsymbol{\varepsilon}) \succ 0$, so that a real, symmetric square root $\mathbf{F}$ exists. Every trajectory actually used for training or testing (Fig. 3) stays at moderate strain magnitudes that satisfy it comfortably, far from this boundary. The homogenized macro-stress used to build this project’s training data and the test-trajectory comparisons of Section 6.3 is *not* the naive volume average of $\{\mathbf{S}_{e,g}\}$ (see the distinction drawn there), but the reaction-force-based, exactly energy-conjugate quantity described there; Section 6.6’s genuine FE² comparison instead matches each row’s own native homogenization output directly against the true FOM’s, for reasons given there. A volume-averaged homogenized *strain* of the converged solution can likewise be computed and differs measurably from $\boldsymbol{\varepsilon}$ for this mixed Dirichlet/free RVE (the free hole boundary is not affine-constrained, so the classical boundary-average identity that would equate the two does not apply), but this paper does not use that quantity anywhere, since $\boldsymbol{\varepsilon}$, being the independent (prescribed) variable rather than a computed output, requires no such choice.

The outer-boundary condition used here is affine Dirichlet, not periodic. Periodic boundary conditions are the more common convention in idealized unit-cell homogenization precisely because, under them, the Hill–Mandel macro-homogeneity condition holds automatically at the level of the deformation gradient, with no further argument needed (Appendix A.3 states the exact identity, and why it does not extend to the strain $\mathbf{E}$ without an added qualification). Neither this identity nor its stress counterpart is automatic for a mixed Dirichlet/free RVE like this one, which is exactly why the applied $\boldsymbol{\varepsilon}$ and the volume-averaged, computed strain/stress quantities are kept explicitly distinct above and in Section 6.3. The reaction-force procedure used there does not rely on periodicity anywhere in its derivation. It follows from the envelope theorem at any converged equilibrium state, for any boundary-condition type, so the same machinery applies unchanged to RVEs for which a periodic idealization is not assumed or not available, such as a finite specimen, an RVE adjacent to a free edge, or a boundary condition dictated by an experimental protocol rather than by periodic tiling.

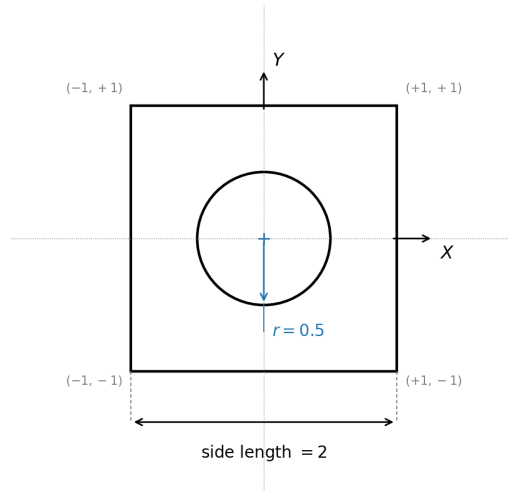


Figure 1: The RVE geometry: a square unit cell of side length 2 (reference coordinates $X, Y \in [-1, 1]$) with a centered circular hole of radius 0.5.

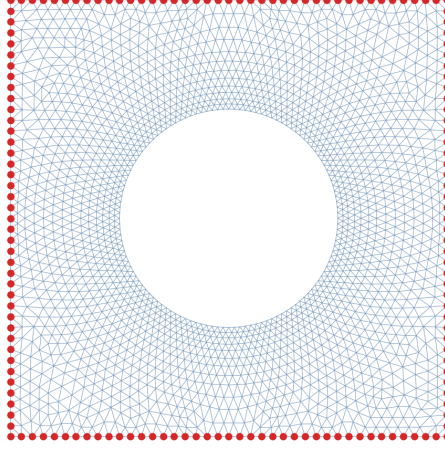


Figure 2: The same geometry as Fig. 1, now meshed with standard displacement-based finite elements. The outer boundary (red) carries the prescribed affine macro-strain Dirichlet condition; the remaining boundary and interior nodes, including the hole boundary, are free. The cell geometry has D_4 (square) symmetry, but the PANN construction in this paper does not assume or exploit it; see Section 4.6.4.

2.2 The offline data-generation campaign: training trajectories and the test path

Every model compared in this paper, the four PANN tiers of Section 4.2.1 as well as the POD basis, the decoder $\mathcal{N}$, and the ECM/MAW–ECM cubature rules behind HPROM–ANN (Section 5), is built from the same underlying dataset. Converged FOM solves of (1) along prescribed macro-strain paths $\varepsilon(t) = (E_{11}(t), E_{22}(t), \gamma_{12}(t))$, each solved state contributing one labeled triple (applied strain, energy-conjugate stress, stored energy) to the dataset. This section describes, once and explicitly, how those paths were chosen and how large the resulting dataset is; Section 6.2 states how it is then split and used to train each model.

Figure 3 shows the full training domain. Ten trajectories form five tension/compression pairs, one pair per swept shear level $\gamma_{12} \in \{0, \pm 0.05, \pm 0.1\}$, the two members of a pair sharing the same shear target but opposite normal-strain branches. Every trajectory is genuinely three-dimensional, not a single planar slice. It first ramps γ_{12} from 0 to its target value at $E_{11} = E_{22} = 0$, then sweeps (E_{11}, E_{22}) in a boustrophedon (zig-zag) pattern within that fixed-shear plane, at a fixed row spacing shared between both branches of a pair, so the sampled states cover an *area* of the (E_{11}, E_{22}) plane rather than a single diagonal ramp. The tension branch of each pair sweeps the larger range $E_{22} \in [0, 2]$; the compression branch sweeps the physically much shorter range $E_{22} \in [-0.1, 0]$ before the hole would close on itself. Sampled at the same fixed row spacing, this asymmetric range is also why each tension branch contributes roughly three times as many solved states as its paired compression branch (16 802–16 842 vs. 5 042–5 082 states per trajectory). Summed over all ten trajectories, the FOM of (1) is solved at 109 460 distinct macro-strain states, the number reused directly in Section 6.2.

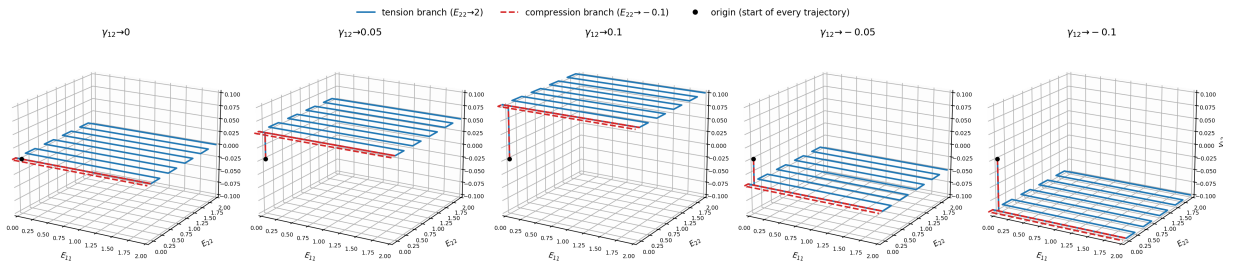


Figure 3: The full three-dimensional macro-strain training domain $(E_{11}, E_{22}, \gamma_{12})$: the ten training trajectories form five tension/compression pairs, one pair per swept shear level $\gamma_{12} \in \{0, \pm 0.05, \pm 0.1\}$, so each panel here fixes γ_{12} at its final value for that pair and shows the (E_{11}, E_{22}) path of both members. Within each panel, the boustrophedon (zig-zag) tension branch sweeps $E_{22} \in [0, 2]$ and the shorter compression branch sweeps $E_{22} \in [-0.1, 0]$, both starting from the origin. FOM solves along all ten paths generate every training snapshot used to build the POD basis, the decoder $\mathcal{N}$, and the ECM/MAW–ECM cubature rules of Section 5.

The test trajectory is a different matter. Every accuracy number reported for every model in this paper (Table 6 onward) is evaluated on a single, separate, 1151-state trajectory that is never included in any training set, referred to throughout simply as the test trajectory. Unlike the ten training trajectories, each of which holds γ_{12} fixed while sweeping (E_{11}, E_{22}) , the test path mixes all three strain components together in a single loading-then-unloading cycle that starts and ends at the origin, reaching $E_{11} \in [-0.05, 1.6]$, $E_{22} \in [-0.05, 1.12]$, $\gamma_{12} \in [-0.075, 0.06]$. It visits tension, compression, and shear simultaneously rather than one at a time, a qualitatively harder generalization test than interpolating within any single training trajectory’s own fixed-shear plane. Figure 4 shows this directly, with a handful of straight probes between extreme mixed states rather than an area sweep at fixed shear.

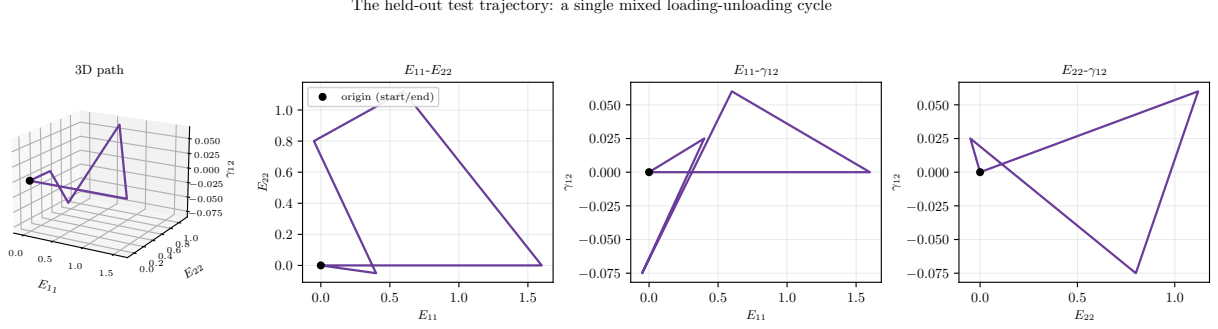


Figure 4: The test trajectory in strain space: one 3D view and three pairwise 2D projections of the same 1151-state path, read directly from the applied-strain sequence used for every reported accuracy number. The path is a small number of straight segments connecting extreme, mixed states and returning to the origin, visually distinct from the area-sweeping training trajectories of Figure 3.

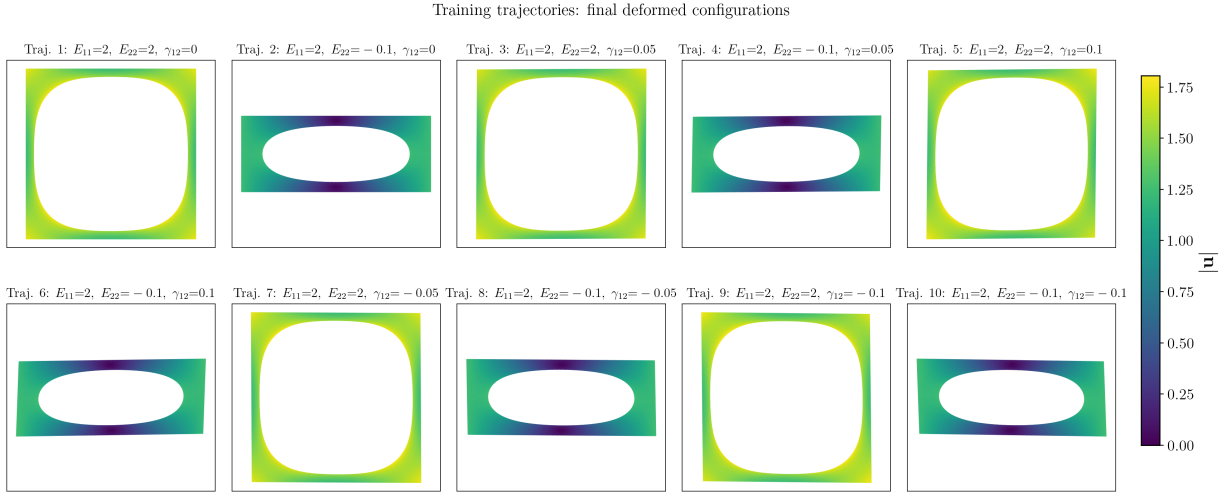


Figure 5: The full displacement field $\mathbf{u} = (\mathbf{u}_d, \mathbf{u}_f)$ at the last (most deformed) configuration of each of the ten training trajectories, reconstructed from the converged FOM solution $\mathbf{u}_f$ (free part) and the prescribed affine Dirichlet data $\mathbf{u}_d$ (boundary part), plotted on the deformed configuration and colored by the nodal displacement magnitude $\|\mathbf{u}\|$ on a single shared color scale across all ten panels (reference coordinates $x, y \in [-1, 1]$). The five tension-branch trajectories ($E_{22} = 2$: 1, 3, 5, 7, 9) show large, roughly equibiaxial bulging with the circular hole rounding toward a square, increasingly skewed by shear; the five compression-branch trajectories ($E_{22} = -0.1$: 2, 4, 6, 8, 10) show a strongly anisotropic, flattened cell with an elliptical hole. The near- D_4 shape visible for the unsheared equibiaxial case (trajectory 1) is a property of this particular deformation state, not an assumption used anywhere in the PANN construction; see Section 4.6.4.

3 The admissibility checklist

This paper organizes six requirements for an admissible hyperelastic surrogate W , with $\boldsymbol{\sigma} = \partial W / \partial \boldsymbol{\varepsilon}$, into a single checklist for clarity of exposition. Each requirement is individually well established and is discussed

with its own citation in the section devoted to it below; the six-item grouping itself is this paper’s own organizing device, not a claim that this exact enumeration is issued by some external authority:

- (C1) **Thermodynamic consistency** (reduced Clausius–Duhem inequality): $\boldsymbol{\sigma}^T \dot{\boldsymbol{\varepsilon}} - \dot{W} \geq 0$.
- (C2) **Objectivity**: $W(\mathbf{Q}\mathbf{F}) = W(\mathbf{F})$ for every $\mathbf{Q} \in \text{SO}(2)$, i.e. every rigid rotation superposed on the current configuration.
- (C3) **Symmetry**: the second Piola–Kirchhoff stress tensor satisfies $\mathbf{S} = \mathbf{S}^T$ (equivalently, in the Voigt notation (4) used throughout, there is no independently parameterized $S_{21} \neq S_{12}$).
- (C4) **Normalization**: $\boldsymbol{\sigma}(\mathbf{0}) = \mathbf{0}$ and $W(\mathbf{0}) = 0$.
- (C5) **Polyconvexity**, in the sense of Ball [2]: see Section 4.6.1 for its precise statement and Section 4.6.2 for its relation to a strictly stronger alternative condition used elsewhere in the literature.
- (C6) **Growth condition**: $W \rightarrow \infty$ as $\boldsymbol{\varepsilon} \rightarrow \infty$ (with extra terms if needed).

Sections 4.2–4.7 (within Section 4) take each point in turn. (C1)–(C4) are satisfied exactly, by construction, independent of training. (C6)’s qualitative coercivity structure (a barrier as $J \rightarrow 0^+$ and super-quadratic growth as any structural stretch diverges) is likewise architectural rather than a trained or fitted behavior, though, as Section 4.7 makes precise, this stops short of the Sobolev-exponent matching a fully rigorous existence proof would additionally require. (C5) requires the most care. Section 4.6 gives Ball’s precise definition, the correct (weaker) necessary condition we actually falsify-test, and the explicit construction that proves our model polyconvex in Ball’s sense; it also shows that a strictly stronger alternative used elsewhere in the literature is not required by polyconvexity and is, for this RVE’s own ground-truth response, false. Section 4.6.4 places this checklist against the current PANN literature as each piece of the construction is introduced: which parts are already correctly and rigorously solved elsewhere, which published claims are imprecise or documentedly incorrect, and which combination of requirements (specifically, a *general* anisotropic material with no assumed symmetry group) remains an open problem that this work targets.

4 Admissibility of the construction: theoretical properties

4.1 Kinematics and stress convention

Let $\mathbf{F}$ be the deformation gradient, and

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}), \quad J = \det \mathbf{F} = \sqrt{\det \mathbf{C}} > 0. \quad (2)$$

The database and the model use the engineering-shear coordinate

$$\boldsymbol{\varepsilon} = \begin{bmatrix} E_{11} \\ E_{22} \\ \gamma_{12} \end{bmatrix}, \quad \gamma_{12} = 2E_{12}, \quad (3)$$

so that the work identity

$$dW = S_{11} dE_{11} + S_{22} dE_{22} + S_{12} d\gamma_{12}, \quad \boldsymbol{\sigma} := \frac{\partial W}{\partial \boldsymbol{\varepsilon}} = \begin{bmatrix} S_{11} \\ S_{22} \\ S_{12} \end{bmatrix} \quad (4)$$

returns exactly the second Piola–Kirchhoff components stored in the direct FOM data, with no factor-two ambiguity in the shear term and no independently trained stress network: every stress reported in this document is $\partial W / \partial \boldsymbol{\varepsilon}$ of the same scalar W .

4.2 (C1) Thermodynamic consistency

For a purely elastic RVE with no internal (dissipative) state variables, the reduced isothermal Clausius–Duhem inequality is $\mathcal{D} := \boldsymbol{\sigma}^T \dot{\boldsymbol{\varepsilon}} - \dot{W} \geq 0$. Since $W = W(\boldsymbol{\varepsilon}(t))$ and W is differentiable,

$$\dot{W} = \frac{\partial W}{\partial \boldsymbol{\varepsilon}} \cdot \dot{\boldsymbol{\varepsilon}} = \boldsymbol{\sigma}^T \dot{\boldsymbol{\varepsilon}} \quad (5)$$

identically, for *any* smooth scalar potential W and *any* strain history $\boldsymbol{\varepsilon}(t)$, by the chain rule alone. Hence $\mathcal{D} \equiv 0 \geq 0$; the inequality holds with equality, and the model is thermodynamically consistent (fully reversible, zero dissipation) for the same reason every hyperelastic (Green-elastic) model is, because the stress is defined as $\partial W / \partial \boldsymbol{\varepsilon}$ in (4) and not as an independently parameterized function. This is not a property that training can violate or that needs to be checked numerically; it is a property of the functional form (4) itself.

4.2.1 A pure-regression baseline: violating (C1) directly

Section 4.2 shows (C1) holds for *any* model of the form $\mathbf{S} = \partial W / \partial \boldsymbol{\varepsilon}$, regardless of which W is learned. It says nothing about a model that is not of this form. As’ad et al. [3] motivate their own hyperelastic-ANN construction by first exhibiting exactly such a model, a plain regression network that maps strain directly to stress with no potential anywhere, as their baseline “regression ANN”, the bottom tier of a three-tier ablation (regression ANN, hyperelastic ANN, convex hyperelastic ANN) that motivates each of their architectural commitments in turn. This section repeats that ablation for this project’s own RVE and its own admissibility checklist, adding a fourth tier below the two certified polyconvex cores and the uncertified free baseline of Section 6.5:

- **Tier 1 (this section):** a pure-regression MLP, widths (128, 128, 64), mapping normalized $[E_{11}, E_{22}, \gamma_{12}]$ straight to normalized $[S_{11}, S_{22}, S_{12}]$. The stress output is a direct forward pass, never a gradient of anything. No energy functional exists in this model at all.
- **Tier 2:** the free hyperelastic PANN of Section 6.5, an unconstrained energy W_θ , so (C1) holds, but nothing else in (C4)–(C6) is guaranteed.
- **Tiers 3a/3b:** the certified polyconvex ICNN and ICKAN of Sections 4.6.4 and 6.4, satisfying all six checklist items exactly.

We can test (C1) directly rather than just argue it. For a genuine hyperelastic model, integrating $\dot{W} = \mathbf{S} \cdot \dot{\boldsymbol{\varepsilon}}$ (Section 4.2) along any closed strain loop $\mathcal{L}$ (a path in strain space with the same start and end point) gives

$$\oint_{\mathcal{L}} \mathbf{S} \cdot d\boldsymbol{\varepsilon} = \oint_{\mathcal{L}} dW = W(\boldsymbol{\varepsilon}_{\text{end}}) - W(\boldsymbol{\varepsilon}_{\text{start}}) = 0, \quad (6)$$

exactly, for *every* closed loop, independently of which admissible W produced $\mathbf{S}$, because this is a direct corollary of (C1), not a new physical assumption. For the tier-1 pure-regression model, $\mathbf{S}$ is not the gradient of anything, so nothing in its construction forces (6), and there is no reason to expect it numerically. We evaluate (6) by trapezoidal quadrature (12800 points per edge, justified below) on four closed strain loops entirely inside the training range (Figure 6): a small axis-aligned normal-strain square, a smooth curve that starts and returns to the undeformed reference state $\boldsymbol{\varepsilon} = \mathbf{0}$ itself while moving through all three strain components at once, a shear loop, and an irregular triangle mixing normal and shear strain, for all four tiers, using each model’s own stress output (a direct forward pass for tier 1, $\partial W_\theta / \partial \boldsymbol{\varepsilon}$ by automatic differentiation for tiers 2–3b).

Table 2: Cyclic work $\oint_{\mathcal{L}} \mathbf{S} \cdot d\boldsymbol{\varepsilon}$ (Pa) around four closed strain loops, trapezoidal quadrature, 12800 points per edge. Exactly zero is expected, by (6), for tiers 2–3b; nothing constrains tier 1. Convergence of this resolution is verified in Figure 7.

Loop	Pure regression (tier 1)	Free (tier 2)	ICNN (tier 3a)	ICKAN (tier 3b)
Small normal square	$-3.041e + 04$ Pa	$4.140e - 01$ Pa	$2.421e - 08$ Pa	$-1.490e - 08$ Pa
Origin-anchored ellipse	$4.225e + 04$ Pa	$4.120e - 01$ Pa	$-6.931e - 06$ Pa	$-3.963e - 02$ Pa
Shear loop	$-3.583e + 04$ Pa	$5.309e - 01$ Pa	$-7.101e - 09$ Pa	$-6.432e - 09$ Pa
Mixed triangle	$9.903e + 03$ Pa	$-7.101e - 01$ Pa	$-1.213e - 02$ Pa	$-1.201e - 02$ Pa

For context, a typical energy density for this RVE (the training-set energy scale used to normalize the loss) is $2.136e + 09$ Pa: the tier-1 residual on the two normal-strain loops and the shear loop is five to six orders of magnitude smaller than this scale (0.002% of it at worst). We verified 12800 points per edge is well converged by refining the quadrature on the mixed triangle (the least regular of the four loops) by $4\times$ at each step, 200 to 12800 points per edge (Figure 7).

The three energy-based tiers shrink at almost exactly the $16\times$ -per-step rate that $\mathcal{O}(h^2)$ trapezoidal error predicts, converging to the true value of zero; the tier-1 residual, having no potential to converge to in the first place, does not move: the signature of a genuine curl in its stress field, not finite-sample noise.

Accuracy does not reveal this violation; if anything, it hides it. Table 6 reports the tier-1 model’s own test-trajectory accuracy: 0.054% overall stress error and 0.033% von Mises error, better than every other tier in this study, including both certified polyconvex cores. This is the sharpest version yet of the point made in Section 6.5 with the free baseline. An aggregate accuracy metric on a test trajectory not only fails to detect the absence of a physical constraint, it can reward the model that discards the most physics, because a direct regression has no architectural constraint competing with the stress fit at all. The cyclic-work test of Table 2 measures something accuracy cannot – it is the only test in this paper that checks (C1) itself, by direct observation rather than as a metric to be optimized.

The four closed strain loops of the (C1) cyclic-work test

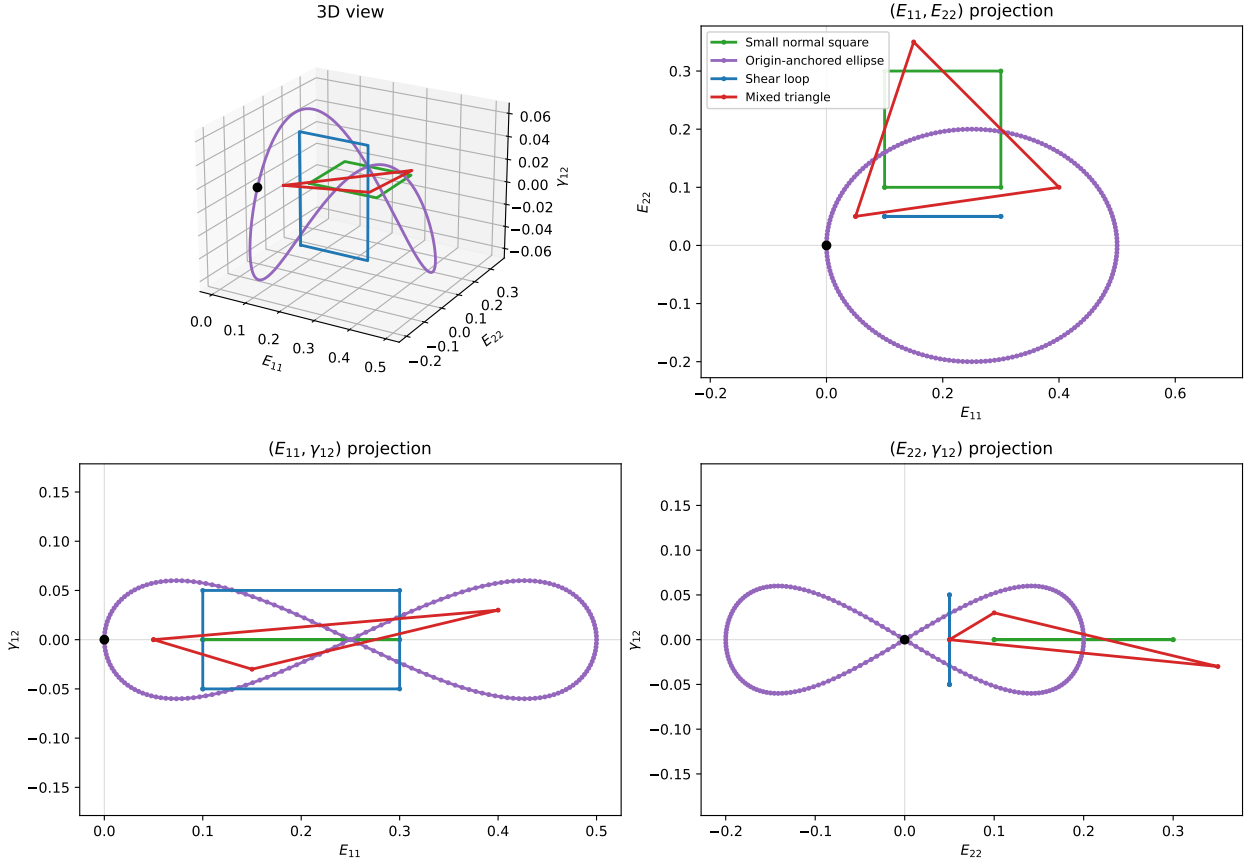


Figure 6: The four closed strain loops evaluated in Table 2: a genuine 3D view (left) and the three pairwise projections (right), so every loop’s shape is visible from some panel without ambiguity. The small square and the shear loop each vary only two of the three strain components (they collapse to a flat segment in whichever projection does not involve their varying components); the origin-anchored ellipse and the mixed triangle both move in all three simultaneously, and only the ellipse starts and returns to the true reference state $\varepsilon = \mathbf{0}$ (the black marker).

4.3 (C2) Objectivity

Objectivity concerns a superposed rigid rotation $\mathbf{Q} \in \text{SO}(2)$ of the *current* configuration, $\mathbf{F} \mapsto \mathbf{Q}\mathbf{F}$. Since

$$(\mathbf{Q}\mathbf{F})^T(\mathbf{Q}\mathbf{F}) = \mathbf{F}^T \mathbf{Q}^T \mathbf{Q} \mathbf{F} = \mathbf{F}^T \mathbf{F} = \mathbf{C}, \quad (7)$$

any scalar built only from $\mathbf{C}$ (equivalently, from $\mathbf{F}^T \mathbf{F}$, $\text{cof}(\mathbf{F})^T \text{cof}(\mathbf{F}) = \text{cof}(\mathbf{C})$, and $\det \mathbf{F}$, as in Section 4.6) automatically satisfies $W(\mathbf{Q}\mathbf{F}) = W(\mathbf{F})$ for every $\mathbf{Q} \in \text{SO}(2)$. Every feature used by the model in Section 4.6 is built exclusively from $\mathbf{C}$ and $\text{cof}(\mathbf{C})$, so objectivity holds identically, not approximately, and not because it was included as a training penalty. This is different from *material* symmetry, which concerns rotating the RVE relative to its own fixed material axes and is discussed next.

4.4 (C3) Symmetry: two different meanings

The word “symmetry” is used for two unrelated properties in continuum mechanics, and the checklist item (C3) refers to the first, not the second:

- **Tensor symmetry**, $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$: a basic consequence of balance of angular momentum in the absence of body couples. It must hold for *any* admissible constitutive model, including anisotropic ones.
- **Material symmetry**: invariance of W under a *material* symmetry group (e.g. the D_4 group of a square lattice), $W(C_{11}, C_{22}, C_{12}) = W(C_{22}, C_{11}, -C_{12})$. This is a *modelling choice*, not a conservation

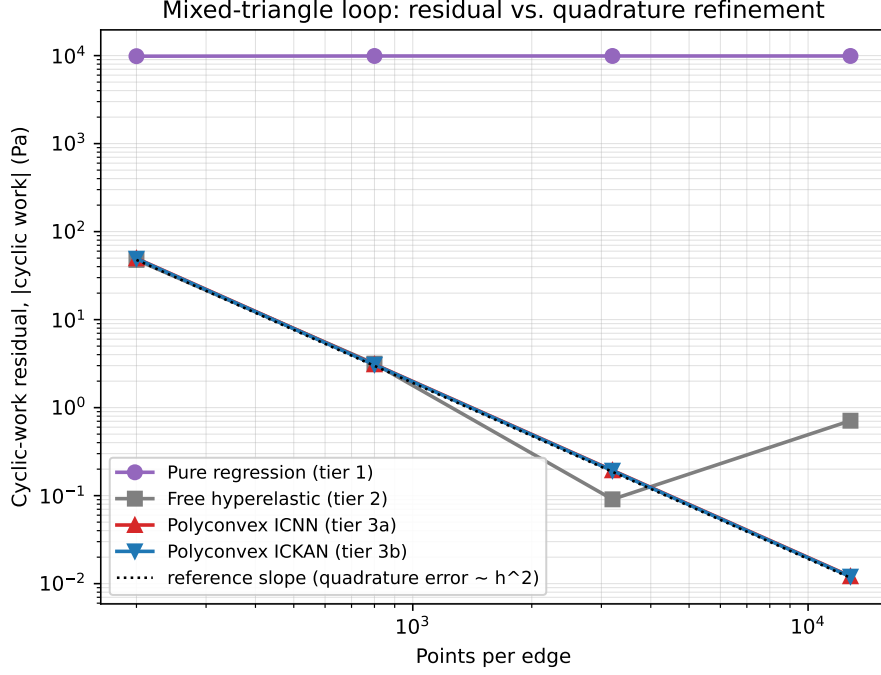


Figure 7: Residual magnitude on the mixed-triangle loop vs. quadrature points per edge, log-log, at the four refinement levels 200, 800, 3200, 12800. The three energy-based tiers fall on the slope expected of $\mathcal{O}(h^2)$ trapezoidal quadrature error (dotted reference line) and are converging to the true value of exactly zero; the tier-1 residual does not move. The free-hyperelastic tier’s residual (tier 2, evaluated in float32) levels off around 10^{-1} –1 Pa at the finest level rather than continuing to shrink, consistent with floating-point precision rather than remaining quadrature error; the two float64 tiers (3a/3b) keep converging cleanly throughout.

law, and this project deliberately does *not* impose it. The RVE is treated as a general anisotropic material, because a square-shaped numerical cell does not by itself make the effective homogenized material isotropic or D_4 symmetric.

Tensor symmetry (the actual content of C3) holds identically in this model. The second Piola stress is parameterized by exactly the three independent numbers (S_{11}, S_{22}, S_{12}) of (4), which are then placed into the symmetric matrix $\begin{pmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{pmatrix}$. There is no separate, independently learned S_{21} that could disagree with S_{12} : $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$ by the structure of the parameterization, not as an approximate or penalized identity. Material symmetry (D_4) is a different, optional property that this model does not claim.

4.5 (C4) Normalization: the stress-free reference state

4.5.1 The energy value: trivial by construction

Let Φ_θ be a scalar network of a feature vector $\mathbf{z}_{\text{pc}}(\mathbf{F})$ defined in Section 4.6, and let $\mathbf{z}_{\text{pc}}(\mathbf{I})$ denote the features evaluated at the reference state $\mathbf{F} = \mathbf{I}$. The energy is defined as

$$W_{\text{pc}}(\mathbf{F}) = \Phi_\theta(\mathbf{z}_{\text{pc}}(\mathbf{F})) - \Phi_\theta(\mathbf{z}_{\text{pc}}(\mathbf{I})) - r \log J + \frac{\beta}{2}(J-1)^2. \quad (8)$$

At $\mathbf{F} = \mathbf{I}$: $\mathbf{z}_{\text{pc}}(\mathbf{F}) = \mathbf{z}_{\text{pc}}(\mathbf{I})$ so the first two terms cancel exactly; $J = 1$ so $\log J = 0$ and $(J-1)^2 = 0$. Hence $W_{\text{pc}}(\mathbf{I}) = 0$ *exactly*, for *any* network Φ_θ and *any* parameters θ , before any training happens. This part of (C4) requires no special argument about the structural directions.

4.5.2 The stress value: not trivial, and this is where the balanced directions matter

Differentiating (8) and evaluating at $\boldsymbol{\varepsilon} = \mathbf{0}$ does *not* automatically give zero, because it requires the gradient of the Φ_θ term to be exactly cancelled by the gradient of the $-r \log J$ term, and the latter only ever points in one fixed, isotropic direction (equal normal parts, zero shear). The argument that this cancellation actually happens has three steps: (i) show that at the reference state, every one of the fifteen features feeding Φ_θ

has a gradient that is a positive multiple of the identity matrix $\mathbf{I}$: an *isotropic* direction, regardless of how anisotropic the feature itself is away from the reference state (Lemma 1 and Proposition 1); (ii) conclude, by the chain rule alone, that Φ_θ 's own gradient at the reference state must then also be a non-negative multiple of $\mathbf{I}$, however Φ_θ was trained; (iii) show that a single scalar r , matched to that multiple, exactly cancels it against $-r \log J$ (Proposition 2). We take these in turn.

Step (i): every feature's reference gradient is isotropic. The fact this step rests on is an elementary property of the 2×2 cofactor applied to a rank-one, unit-norm outer product: it turns the direction into its own orthogonal complement.

Lemma 1 (Isotropic reference derivative). *Let $\mathbf{d}$ be a unit vector in $\mathbb{R}^2$. Then $\text{cof}(\mathbf{d} \otimes \mathbf{d}) = \mathbf{I} - \mathbf{d} \otimes \mathbf{d}$.*

Proof. Write $\mathbf{d} = (\cos \theta, \sin \theta)$, so that

$$\mathbf{d} \otimes \mathbf{d} = \begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{pmatrix}. \quad (9)$$

The 2×2 cofactor formula (swap the diagonal entries, negate the off-diagonal ones) gives

$$\text{cof}(\mathbf{d} \otimes \mathbf{d}) = \begin{pmatrix} \sin^2 \theta & -\cos \theta \sin \theta \\ -\cos \theta \sin \theta & \cos^2 \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{pmatrix} = \mathbf{I} - \mathbf{d} \otimes \mathbf{d}, \quad (10)$$

using $\cos^2 \theta + \sin^2 \theta = 1$ on the diagonal entries. $\square$

With this in hand, the same conclusion extends from a single direction to every one of the fifteen features in (29), provided the directions within each system are balanced:

Proposition 1 (Every feature has an isotropic $\mathbf{C}$ -gradient at $\mathbf{C} = \mathbf{I}$). *Let $\mathbf{d}_{k1}, \mathbf{d}_{k2}, \mathbf{d}_{k3}$ be unit vectors and $w_{k1}, w_{k2}, w_{k3} > 0$ satisfy the balance condition $\sum_i w_{ki} \mathbf{d}_{ki} \otimes \mathbf{d}_{ki} = \mathbf{I}$ for each system $k = 1, 2, 3$, and recall the fifteen features of Section 4.6.4: $I_1 = \text{tr } \mathbf{C}$, and, for $p = 2, 3$,*

$$Q_{k,p}^F = \sum_i w_{ki} (\mathbf{d}_{ki} \cdot \mathbf{C} \mathbf{d}_{ki})^p, \quad Q_{k,p}^H = \sum_i w_{ki} (\mathbf{d}_{ki} \cdot \text{cof}(\mathbf{C}) \mathbf{d}_{ki})^p. \quad (11)$$

Then at $\mathbf{C} = \mathbf{I}$, every one of these thirteen scalars (I_1 and twelve Q 's, three systems, two powers, two blocks), together with J and J^2 , has a $\mathbf{C}$ -gradient proportional to the identity: $dz_m/d\mathbf{C}|_{\mathbf{C}=\mathbf{I}} = c_m \mathbf{I}$ for every feature z_m , with the explicit coefficients c_m collected in Table 3.

Proof. Taking the trace of the balance identity and using $|\mathbf{d}_{ki}| = 1$ gives $\sum_i w_{ki} |\mathbf{d}_{ki}|^2 = \text{tr}(\mathbf{I}) = 2$, i.e. $\sum_i w_{ki} = 2$ for every system k (consistent with the numerical weights of Table 4). This one identity, together with Lemma 1, is all four cases below need.

- **The invariant I_1 .** Trivially, $dI_1/d\mathbf{C} = \mathbf{I}$ for every $\mathbf{C}$, so in particular at $\mathbf{C} = \mathbf{I}$.
- **The F -block features $Q_{k,p}^F$.** By the product/chain rule,

$$\frac{dQ_{k,p}^F}{d\mathbf{C}} = \sum_i p w_{ki} (\mathbf{d}_{ki} \cdot \mathbf{C} \mathbf{d}_{ki})^{p-1} \mathbf{d}_{ki} \otimes \mathbf{d}_{ki}. \quad (12)$$

At $\mathbf{C} = \mathbf{I}$, each factor $\mathbf{d}_{ki} \cdot \mathbf{I} \mathbf{d}_{ki} = |\mathbf{d}_{ki}|^2$ equals 1, so the $(p-1)$ -power drops out and only the balance sum survives:

$$\left. \frac{dQ_{k,p}^F}{d\mathbf{C}} \right|_{\mathbf{C}=\mathbf{I}} = p \sum_i w_{ki} \mathbf{d}_{ki} \otimes \mathbf{d}_{ki} = p \mathbf{I}. \quad (13)$$

- **The $\text{cof}(\mathbf{C})$ -block features $Q_{k,p}^H$.** This case needs one extra observation before the same trick applies: $\mathbf{d} \cdot \text{cof}(\mathbf{C}) \mathbf{d}$ is itself *linear* in $\mathbf{C}$, because $\mathbf{d} \cdot \text{cof}(\mathbf{C}) \mathbf{d} = \text{cof}(\mathbf{C}) : (\mathbf{d} \otimes \mathbf{d})$ and the cofactor operation is self-adjoint under the Frobenius inner product for 2×2 matrices ($A : \text{cof}(B) = B : \text{cof}(A)$, checked by direct expansion), so $\mathbf{d} \cdot \text{cof}(\mathbf{C}) \mathbf{d} = \mathbf{C} : \text{cof}(\mathbf{d} \otimes \mathbf{d})$: linear in $\mathbf{C}$, with the *constant* gradient $\text{cof}(\mathbf{d} \otimes \mathbf{d}) = \mathbf{I} - \mathbf{d} \otimes \mathbf{d}$ supplied by Lemma 1. Differentiating as before,

$$\frac{dQ_{k,p}^H}{d\mathbf{C}} = \sum_i p w_{ki} (\mathbf{d}_{ki} \cdot \text{cof}(\mathbf{C}) \mathbf{d}_{ki})^{p-1} (\mathbf{I} - \mathbf{d}_{ki} \otimes \mathbf{d}_{ki}). \quad (14)$$

At $\mathbf{C} = \mathbf{I}$, $\text{cof}(\mathbf{I}) = \mathbf{I}$, so $\mathbf{d}_{ki} \cdot \text{cof}(\mathbf{I})\mathbf{d}_{ki} = 1$ and the $(p-1)$ -power again drops out, leaving

$$\left. \frac{dQ_{k,p}^H}{d\mathbf{C}} \right|_{\mathbf{C}=\mathbf{I}} = p \left[\left(\sum_i w_{ki} \right) \mathbf{I} - \sum_i w_{ki} \mathbf{d}_{ki} \otimes \mathbf{d}_{ki} \right] = p(2\mathbf{I} - \mathbf{I}) = p\mathbf{I}, \quad (15)$$

using $\sum_i w_{ki} = 2$ and the balance identity a second time.

- **The volumetric terms J, J^2 .** The standard determinant-derivative identity gives

$$\frac{dJ}{d\mathbf{C}} = \frac{1}{2J} \text{cof}(\mathbf{C}), \quad \left. \frac{dJ}{d\mathbf{C}} \right|_{\mathbf{C}=\mathbf{I}} = \frac{1}{2}\mathbf{I}; \quad \frac{dJ^2}{d\mathbf{C}} = 2J \frac{dJ}{d\mathbf{C}} = \text{cof}(\mathbf{C}), \quad \left. \frac{dJ^2}{d\mathbf{C}} \right|_{\mathbf{C}=\mathbf{I}} = \mathbf{I}. \quad (16)$$

□

Table 3: Isotropic-derivative coefficient c_m (i.e. $dz_m/d\mathbf{C}|_{\mathbf{C}=\mathbf{I}} = c_m \mathbf{I}$) for each of the 15 features, from Proposition 1.

Feature	I_1	$Q_{k,2}^F, Q_{k,2}^H$ (each of 3 systems)	$Q_{k,3}^F, Q_{k,3}^H$ (each of 3 systems)	J	J^2
c_m	1	2	3	1/2	1

In words: every feature’s gradient at the reference state points in the same isotropic direction $\mathbf{I}$; only the multiple c_m changes from feature to feature. This is exactly the fact Step (ii) needs.

Steps (ii)-(iii): the multiple cancels exactly against the volumetric barrier. By the chain rule, Φ_θ ’s own reference gradient is a weighted sum of the fifteen feature gradients just computed, each of which is a multiple of $\mathbf{I}$, so the sum is too, automatically, for *any* trained Φ_θ . Matching that multiple against the barrier term’s own (also isotropic) reference gradient is what Proposition 2 makes precise.

Proposition 2 (Exact cancellation of the reference stress). *Define*

$$r := \frac{1}{2} \left(\frac{\partial \Phi_\theta}{\partial E_{11}} + \frac{\partial \Phi_\theta}{\partial E_{22}} \right) \Big|_{\varepsilon=0}. \quad (17)$$

Then $\boldsymbol{\sigma}(\mathbf{0}) = \mathbf{0}$ exactly.

Proof. 1. Φ_θ ’s reference gradient is $2\lambda\mathbf{I}$ for some $\lambda \geq 0$. By the chain rule and Proposition 1,

$$\left. \frac{d\Phi_\theta}{d\mathbf{C}} \right|_{\mathbf{C}=\mathbf{I}} = \sum_m \left. \frac{\partial \Phi_\theta}{\partial z_m} \right|_{\mathbf{z}(\mathbf{I})} \left. \frac{dz_m}{d\mathbf{C}} \right|_{\mathbf{C}=\mathbf{I}} = \left(\sum_m c_m \left. \frac{\partial \Phi_\theta}{\partial z_m} \right|_{\mathbf{z}(\mathbf{I})} \right) \mathbf{I} =: \lambda \mathbf{I}. \quad (18)$$

Because Φ_θ is an ICNN with non-negative weights at every layer (Section 4.6.4), every partial derivative $\partial \Phi_\theta / \partial z_m$ is non-negative; combined with $c_m > 0$ from Table 3, λ is a non-negative combination of non-negative terms, so $\lambda \geq 0$ automatically, independent of training, for any θ . Converting from $\mathbf{C}$ to $\mathbf{E}$ via $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ doubles the gradient, $d\Phi_\theta/d\mathbf{E}|_{\mathbf{E}=0} = 2\lambda\mathbf{I}$, which in the engineering-strain coordinate (3) reads

$$\left. \frac{\partial \Phi_\theta}{\partial \boldsymbol{\varepsilon}} \right|_{\mathbf{0}} = (2\lambda, 2\lambda, 0)^T \quad (19)$$

(the shear entry vanishes because it is the off-diagonal entry of the isotropic matrix $2\lambda\mathbf{I}$).

2. **The barrier term’s reference gradient has the same isotropic shape.** Since $\log J = \frac{1}{2} \log \det \mathbf{C}$,

$$\frac{d(\log J)}{d\mathbf{C}} = \frac{1}{2 \det \mathbf{C}} \text{cof}(\mathbf{C}), \quad \left. \frac{d(\log J)}{d\mathbf{C}} \right|_{\mathbf{C}=\mathbf{I}} = \frac{1}{2}\mathbf{I}; \quad (20)$$

converting to $\mathbf{E}$ the same way as in Step 1 gives $\partial(\log J)/\partial \boldsymbol{\varepsilon}|_{\mathbf{0}} = (1, 1, 0)^T$: the same isotropic direction as Step 1’s result, by construction.

3. **Match the two and cancel.** By definition (17) and Step 1, $r = \frac{1}{2}(2\lambda + 2\lambda) = 2\lambda$: r is defined precisely so that $r(1, 1, 0)^T$ equals Φ_θ ’s own reference gradient $(2\lambda, 2\lambda, 0)^T$ from Step 1. Summing all three terms of (8) at $\boldsymbol{\varepsilon} = \mathbf{0}$ then gives

$$\boldsymbol{\sigma}(\mathbf{0}) = \underbrace{(2\lambda, 2\lambda, 0)^T}_{\text{from } \Phi_\theta} - r \underbrace{(1, 1, 0)^T}_{\text{from } -r \log J} + \underbrace{\beta(J-1)\nabla J|_{J=1}}_{=0, \text{ since } J=1} = \mathbf{0}. \quad (21)$$

□

Remark 1. The balance condition $\sum_i w_{ki} \mathbf{d}_{ki} \otimes \mathbf{d}_{ki} = \mathbf{I}$ is what makes this possible with a single scalar r : without it, the reference gradient of Φ_θ would generally be an anisotropic 3-vector (unequal normal components, nonzero shear), and no scalar multiple of the purely isotropic direction $(1, 1, 0)^T$ could cancel it exactly. The construction would then leave a spurious residual stress at the undeformed configuration: a real defect, unlike the SPD-tangent distinction drawn in Section 4.6, which does not indicate any defect of the FOM at all.

4.6 (C5) Polyconvexity

4.6.1 Definition

Polyconvexity (Ball [2]) is a condition on the *functional form* of W as a function of $\mathbf{F}$, namely that there exists a jointly convex function G of three *independent* block arguments such that

$$W(\mathbf{F}) = G(\mathbf{F}, \text{cof } \mathbf{F}, \det \mathbf{F}), \quad G \text{ convex on } \mathbb{R}^{2 \times 2} \times \mathbb{R}^{2 \times 2} \times \mathbb{R}. \quad (22)$$

This is an existence-theory condition; Ball used it (together with suitable growth, Section 4.7) to prove, via the direct method of the calculus of variations, that minimizers of the total elastic energy exist. It says nothing by itself about the sign of $\partial \boldsymbol{\sigma} / \partial \boldsymbol{\varepsilon}$ at a given deformation state. The correct hierarchy (Ball [2]; see also Dacorogna’s standard reference on the calculus of variations) is

$$\text{convex} \implies \text{polyconvex} \implies \text{quasiconvex} \implies \text{rank-one convex}, \quad (23)$$

with each arrow strict in general; Section 4.6.4 proves our construction satisfies (22) directly, and Section 4.6.3 gives the weakest, pointwise-checkable necessary consequence of it that we verify numerically.

4.6.2 A strictly stronger alternative used elsewhere, and why it is not adopted here

Polyconvexity (22) is not the only architectural route to a well-posed hyperelastic model, and it is worth being precise about how it relates to a different, strictly stronger condition used elsewhere in the literature: a globally symmetric positive definite (SPD) tangent, $\mathbf{D} = \partial \boldsymbol{\sigma} / \partial \boldsymbol{\varepsilon} \succ 0$ everywhere. As’ad, Avery and Farhat [3] adopt exactly this stronger condition for their own anisotropic fabric RVE, built with the same FE² pipeline used here and validated inside an actual coupled fluid–structure simulation (the supersonic inflation of a parachute canopy, the only downstream robustness demonstration surveyed in this paper), for a specific, well-documented reason, in their own words: “enforcing polyconvexity in $\mathbf{F}$ is difficult as $\mathbf{E}$ is not in general convex in $\mathbf{F}$.” Gao, Neff, Roventa and Thiel [26] likewise build a valid global-minimizer existence theorem directly on convexity of W in $\mathbf{C}$, which the same reparametrization $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ shows is equivalent to this SPD-tangent condition, and the same condition, enforced by an SPD-NN predicting the tangent’s Cholesky factor directly rather than an ICNN, underlies Xu, Huang and Darve [27] for path-dependent elasto-plastic anisotropic response. Both are legitimate, coherent design choices, not errors, and As’ad et al.’s choice buys real robustness in their finite-element examples (Newton’s method diverges for a non-convex ANN under load, converges for their convex-in- $\mathbf{E}$ one), but neither is polyconvexity. Ball’s existence theorem needs only (22) together with growth, not a globally SPD tangent, so the two conditions are logically independent, and the SPD-tangent route remains available only because, unlike plain convexity of W in $\mathbf{F}$ itself, convexity in $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ raises no contradiction with objectivity (objectivity forces W to be constant along the curved, non-convex orbit $\{\mathbf{Q}\mathbf{F} : \mathbf{Q} \in \text{SO}(2)\}$, which for $n = 2$ includes $\mathbf{Q} = -\mathbf{I}$ and would force convexity in $\mathbf{F}$ to control W along a segment through the singular matrix $\frac{1}{2}(\mathbf{F} + (-\mathbf{F})) = 0$; convexity in the already-objective $\mathbf{C}$ has no such obstruction).

For this RVE specifically, however, the SPD-tangent route is not available. The independent FOM tangent audit of this project measures $\mathbf{D} = \partial \mathbf{S} / \partial \boldsymbol{\varepsilon}$ by finite differences at several deformation states, including a test-trajectory state where the *smallest eigenvalue of $\mathbf{D}$* is -0.0648 GPa, i.e. negative. The true, homogenized RVE response is therefore not SPD-tangent everywhere (a signature of an incremental-stability transition in the underlying microstructure, not a modelling artifact), and a surrogate forced to satisfy $\mathbf{D} \succ 0$ globally would necessarily *contradict* the FOM at that state. We therefore target the correct, weaker, and for this RVE actually achievable condition, polyconvexity, and deliberately do not impose

$$\frac{\partial \mathbf{S}}{\partial \boldsymbol{\varepsilon}} \succ 0 \quad (\text{not imposed, and not implied by (C5)}). \quad (24)$$

4.6.3 The necessary condition we verify: rank-one convexity

The weakest property in the hierarchy (23), rank-one convexity, equivalent to the Legendre–Hadamard (ellipticity) condition, is the correct *necessary* consequence of polyconvexity that can be checked pointwise:

$$\left. \frac{d^2}{dt^2} \right|_{t=0} W(\mathbf{F} + t \mathbf{a} \otimes \mathbf{b}) \geq 0 \quad \text{for all } \mathbf{F} \text{ in the domain and all } \mathbf{a}, \mathbf{b} \in \mathbb{R}^2. \quad (25)$$

This is a condition on the curvature of W *restricted to rank-one lines in $\mathbf{F}$ -space*, which is a small, highly structured subset of directions, not on the full 3×3 Hessian $\mathbf{D}$ in strain space required (and false) in Section 4.6.2. Rank-one convexity failing is the classical mathematical signature of loss of ellipticity (shear banding, microstructure buckling): a real, physically meaningful phenomenon that a good model should be able to reproduce, not one it should be forced to exclude. (25) is what the evaluator in this project samples: 64 random pairs $(\mathbf{F}, \mathbf{a}, \mathbf{b})$ with $\det \mathbf{F} > 0$, giving a smallest observed value of $5.395e-03$ GPa. Being positive is consistent with (necessary for) polyconvexity; it is a numerical falsification attempt, not the proof, which is given next.

4.6.4 The construction, and the proof that it is polyconvex

For each $k = 1, 2, 3$, fix three unit structural directions $\mathbf{d}_{ki}$ and positive weights w_{ki} satisfying the balance condition

$$\sum_{i=1}^3 w_{ki} \mathbf{d}_{ki} \otimes \mathbf{d}_{ki} = \mathbf{I}, \quad (26)$$

with the three direction systems (angles in degrees) and weights recorded in Table 4, chosen without reference to any material symmetry group; Section 4.6.4 gives the explicit, reproducible procedure that produced them. A symmetric, direction-built structural tensor entering the energy alongside the classical isotropic invariants is a classical device, not one introduced here, since it traces to Boehler [31] and Spencer [32, 33]. A discrete set of weighted directions satisfying an exactness condition analogous to (26), there for scalar polynomial quadrature on the sphere rather than for this tensor identity, is independently used by Bažant and Oh [34] for numerical integration.

This is deliberately not tied to any named symmetry group, which is worth placing against how the two closest prior polyconvex constructions handle the same requirement. Klein et al. [1] (the primary reference for the correct polyconvexity definition used throughout Section 4.6) give two ICNN constructions restricted a priori to a symmetry group $\mathcal{G} \subset SO(3)$ (cubic, transversely isotropic) via a structural tensor. Their invariant-based model W^I gets objectivity, material symmetry, and polyconvexity exactly, but not the reference state, and they say why directly: an exact stress-free correction “preserves polyconvexity when formulated in terms of $\mathbf{F}$ instead of $\mathbf{C}$, [but] is not compatible with the methods of incorporating objectivity and material symmetry which are used in this work”; their deformation-gradient model W^F trades the opposite way, exact reference state and polyconvexity, but objectivity only approximated by data augmentation over 1024 random rotations. Linden et al. [20] close this specific gap for isotropic and transversely isotropic materials, achieving all four properties *simultaneously* within a fixed structural-tensor symmetry class, but, as already noted in Section 1, do not extend further. Klein et al. [24] widen the named-group set considerably (cubic, tetragonal, ...) via a triclinic integrity basis of $\mathbf{C}, \text{cof}(\mathbf{C}), J$ restricted by an explicit group-symmetrization operator, though every worked example (their Section 5) still uses the cubic group specifically, and raise, without developing it into a proof or example, that the same model “could also be employed without incorporating material symmetry at all” (Appendix A.2 examines what their own machinery would already imply in this limit, and why we do not credit it as an achieved result). This is the open problem this paper answers for a 2D RVE with no assumed symmetry class. The balanced, non- D_4 -closed direction systems of Table 4 replace the structural tensor and are not tied to any symmetry group, while Proposition 1 still supplies the isotropic reference-derivative structure the correction needs, giving, from here onward, an explicit statement of the general-material case and a proof against (C1)–(C6) not yet carried out anywhere in this literature, complementing rather than competing with Klein et al.’s differently-scoped result for named symmetry groups.

How the direction systems are chosen: an explicit, reproducible procedure. Fixing three angles $\theta_{k1}, \theta_{k2}, \theta_{k3} \in [0^\circ, 180^\circ)$ for a candidate system k determines the unit directions $\mathbf{d}_{ki} = (\cos \theta_{ki}, \sin \theta_{ki})$; the balance condition (26) is then three scalar equations for the three unknown weights w_{k1}, w_{k2}, w_{k3} , obtained by equating each of the three independent entries of the symmetric 2×2 matrix $\sum_i w_{ki} \mathbf{d}_{ki} \otimes \mathbf{d}_{ki}$ to the

Table 4: The three structural direction systems and their positive weights, each satisfying (26) to machine precision (maximum residual 2.2×10^{-16}).

System k	Angles $\theta_{k1}, \theta_{k2}, \theta_{k3}$	Weights w_{k1}, w_{k2}, w_{k3}
1	$0^\circ, 50^\circ, 130^\circ$	0.2959, 0.8520, 0.8520
2	$20^\circ, 83^\circ, 151^\circ$	0.5571, 0.7941, 0.6488
3	$12^\circ, 76^\circ, 143^\circ$	0.5760, 0.7930, 0.6310

corresponding entry of $\mathbf{I}$:

$$\underbrace{\begin{pmatrix} \cos^2 \theta_{k1} & \cos^2 \theta_{k2} & \cos^2 \theta_{k3} \\ \sin^2 \theta_{k1} & \sin^2 \theta_{k2} & \sin^2 \theta_{k3} \\ \sin \theta_{k1} \cos \theta_{k1} & \sin \theta_{k2} \cos \theta_{k2} & \sin \theta_{k3} \cos \theta_{k3} \end{pmatrix}}_{=: A(\theta_{k1}, \theta_{k2}, \theta_{k3})} \begin{pmatrix} w_{k1} \\ w_{k2} \\ w_{k3} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}. \quad (27)$$

For a generic angle triple, A is invertible and (27) has a unique solution $\mathbf{w}_k$ from a single 3×3 linear solve, but that solution is not guaranteed to be admissible, because the construction additionally needs $w_{ki} > 0$ (used explicitly in the proof of Proposition 3, where positivity of the weights is what lets a sum of convex terms stay convex), and nothing in (27) forbids a negative or zero entry for a poorly chosen triple. The systems of Table 4 were produced by the direct, mechanical procedure this makes possible: sample a candidate angle triple, solve (27) for $\mathbf{w}_k$, and keep the triple only if A is well-conditioned and every solved weight is strictly positive; repeat until three sufficiently distinct systems are found. The three reported systems are well-conditioned in this sense ($\text{cond}(A) \approx 1.9, 2.3, 2.2$ for systems 1–3 respectively) and their solved weights agree with Table 4 to the reported precision.

Two honest limits of this procedure are stated plainly here rather than left implicit. First, none of the proofs in this section use any property of these specific nine angles beyond (26) and $w_{ki} > 0$: Propositions 1 and 3 hold verbatim for *any* three systems produced by the same accept/reject procedure, so the particular numbers in Table 4 are a convenient, verified choice, not a load-bearing one. Second, and more fundamentally, this paper does *not* prove that the resulting map from $\mathbf{C}$ to the 15-dimensional feature vector (29) is injective (that two different admissible $\mathbf{C}$ can never produce identical features), nor that three direction systems are the minimum or optimal number for this construction. A single system already supplies an admissible, polyconvex model, and more than three would remain admissible by the identical argument. Three systems were adopted because, empirically, they gave the anisotropic expressivity needed to fit this RVE’s own test-trajectory response (Section 6.3) with a single fixed feature set, not because of a representational-completeness argument for a general anisotropic material. This is a reasonable, structured construction consistent with every proof and every numerical result in this paper, not a theorem of representational optimality (Section 7 returns to this point).

From these we form

$$\begin{aligned} I_1 &= \text{tr } \mathbf{C} = |\mathbf{F}|^2, \\ Q_{k,p}^F &= \sum_{i=1}^3 w_{ki} (\mathbf{d}_{ki} \cdot \mathbf{C} \mathbf{d}_{ki})^p = \sum_{i=1}^3 w_{ki} |\mathbf{F} \mathbf{d}_{ki}|^{2p}, \quad p = 2, 3, \\ Q_{k,p}^H &= \sum_{i=1}^3 w_{ki} (\mathbf{d}_{ki} \cdot \text{cof}(\mathbf{C}) \mathbf{d}_{ki})^p = \sum_{i=1}^3 w_{ki} |\text{cof}(\mathbf{F}) \mathbf{d}_{ki}|^{2p}, \quad p = 2, 3, \end{aligned} \quad (28)$$

using $\mathbf{C} : \text{cof}(A \otimes A)$ identities as in Lemma 1 to recognize $\mathbf{d} \cdot \text{cof}(\mathbf{C}) \mathbf{d} = |\text{cof}(\mathbf{F}) \mathbf{d}|^2$. The evaluated feature vector is

$$\mathbf{z}_{\text{pc}} = (I_1, \{Q_{k,2}^F, Q_{k,3}^F, Q_{k,2}^H, Q_{k,3}^H\}_{k=1}^3, J, J^2) \in \mathbb{R}^{15}. \quad (29)$$

For a single direction and $p = 1$, $Q_{k,1}^F$ reduces to the classical pseudo-invariant $\mathbf{a} \cdot \mathbf{C} \mathbf{a}$ used throughout anisotropic hyperelasticity [32, 33, 31]; Schröder and Neff [35] prove polyconvexity of exactly this invariant, and of its $\text{cof}(\mathbf{C})$ -based counterpart, raised to any power for a single fixed direction, which Proposition 3 below extends to balanced sums over multiple directions. Gasser, Ogden and Holzapfel [36] generalize the single-direction invariant to a continuous integral over a fiber-orientation density, the continuous analogue of the discrete weighted sum used here.

The sixth-power terms are not cosmetic. In two dimensions a balanced quartic aggregate alone obeys an accidental 90° identity: writing $t_i = \mathbf{d}_i \cdot \mathbf{C} \mathbf{d}_i$, the 90° -rotated directions have $t'_i = \text{tr}(\mathbf{C}) - t_i$ (true for *any* unit vector and its orthogonal complement, by trace decomposition in an orthonormal basis), and the

balance identity gives $\sum_i w_i (\text{tr}(\mathbf{C}) - t_i)^2 = \sum_i w_i t_i^2$ exactly (expand the square and use $\sum_i w_i t_i = \text{tr}(\mathbf{C})$ and $\sum_i w_i = 2$). The cubic aggregate $\sum_i w_i t_i^3$ does not obey this identity in general, so it supplies genuine anisotropic information while (Proposition 1) its first derivative at $\mathbf{C} = \mathbf{I}$ remains isotropic.

Remark 2 (The fifteen features are not algebraically independent). *The Q^H features are, in fact, exact algebraic functions of I_1 and the Q^F features, so the fifteen entries of $\mathbf{z}_{\text{pc}}$ in (29) are not fifteen independent scalars. This does not weaken the construction (Propositions 1 and 3 hold regardless), but the two cases are not equally justified. $Q_{k,3}^H$ earns its own network input for a concrete architectural reason: its exact reconstruction from $I_1, Q_{k,2}^F, Q_{k,3}^F$ requires negative coefficients that a monotone (non-negative-weight) ICNN cannot represent internally, so supplying it directly gives the network access to combinations it could not otherwise reach. $Q_{k,2}^H$ has no such justification: the identity $Q_{k,2}^H = Q_{k,2}^F$ (an exact 2D consequence of $\text{cof } \mathbf{C} = \text{tr}(\mathbf{C})\mathbf{I} - \mathbf{C}$, verified to machine precision for the trained model) has coefficient +1, which a monotone network can already represent by weighting $Q_{k,2}^F$ alone; $Q_{k,2}^H$ therefore adds no expressiveness in this specific 2D construction. We retain it anyway, for uniformity with the general F/H recipe (built from $\text{cof } \mathbf{F}$ exactly as Q^F is built from $\mathbf{F}$, for every k and every p), which is dimension-independent and does not exhibit this collapse in 3D, where $\text{cof } \mathbf{C}$ is a genuinely quadratic (not affine) function of $\mathbf{C}$. Appendix A.1 gives the exact identities and this reasoning in full.*

Let $\Phi_\theta : \mathbb{R}^{15} \rightarrow \mathbb{R}$ be an input-convex neural network (ICNN, Amos et al. [37]) with Softplus activations and *every* weight matrix (input-to-hidden at every layer, hidden-to-hidden, and hidden-to-output) reparameterized as $\text{softplus}(\cdot) + \varepsilon_0 > 0$. Non-negativity at every layer (not just hidden-to-hidden, as in the original ICNN convexity-only construction) is what additionally makes Φ_θ *coordinate-wise non-decreasing* in $\mathbf{z}_{\text{pc}}$, a property used explicitly below. The energy is (8), repeated here with $r, \beta > 0$:

$$W_{\text{pc}}(\mathbf{F}) = \Phi_\theta(\mathbf{z}_{\text{pc}}) - \Phi_\theta(\mathbf{z}_{\text{pc}}(\mathbf{I})) - r \log J + \frac{\beta}{2}(J - 1)^2. \quad (8)$$

Figure 8 collects this whole construction end to end: the deformation gradient, the 15-dimensional feature map, the input-convex architecture itself (widths 15-64-64-32-1, Softplus activations, every weight non-negative, with the characteristic ICNN skip connections feeding $\mathbf{z}_{\text{pc}}$ directly into every layer rather than only the first), the resulting polyconvex energy, and the stress obtained from it by automatic differentiation.

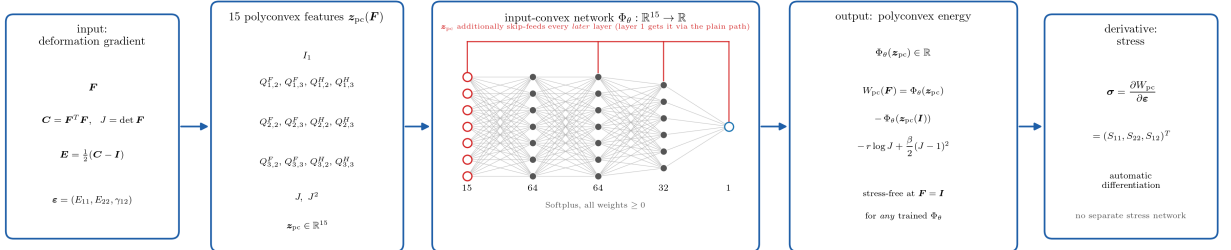


Figure 8: The polyconvex ICNN core end to end: deformation gradient $\mathbf{F}$, the 15-dimensional feature map $\mathbf{z}_{\text{pc}}(\mathbf{F})$ of (29), the input-convex network Φ_θ (hidden widths (64, 64, 32) between the 15-dimensional feature layer and the scalar output, Softplus activations, every weight matrix non-negative; only a representative subset of units per layer is drawn), the resulting polyconvex energy (8), and the stress $\boldsymbol{\sigma} = \partial W_{\text{pc}} / \partial \boldsymbol{\varepsilon}$ obtained from it by automatic differentiation, with no independently parameterized stress network.

The literature documents two concrete, checkable errors of exactly this kind, useful context for how easy this is to get wrong even with a stated intent to be rigorous. Klein et al. [1] report that Ghaderi et al. [30] claim a polyconvex ANN potential while using hyperbolic-tangent activations, which are not convex; we cross-checked this against the original, whose own Section 4 confirms the network core uses “soft plus . . . , sinusoid . . . and hyperbolic tangent” activations together, none of which is convex and non-decreasing throughout. They report the same failure mode for Tac et al. (2021), whose model uses the non-polyconvex invariant $\bar{I}_2 = \frac{1}{2}(\text{tr } \bar{\mathbf{C}})^2 - \frac{1}{2} \text{tr } \bar{\mathbf{C}}^2$ as a direct input (Hartmann and Neff, cited by Klein et al. as their Lemma 2.4). Both illustrate the same point this construction is built to avoid. Polyconvexity is a property of the specific

functional form all the way down to the activation function and the input invariants, not a label attachable to an architecture in general.

Proposition 3 (Polyconvexity of W_{pc}). $W_{\text{pc}}(\mathbf{F}) = G(\mathbf{F}, \text{cof } \mathbf{F}, J)$ for a function G that is jointly convex on $\mathbb{R}^{2 \times 2} \times \mathbb{R}^{2 \times 2} \times (0, \infty)$.

Proof. The argument has four stages: each of the fifteen features is convex in its own block variable; a function of only one block of a convex, block-separated feature is therefore convex in the joint variable $(\mathbf{F}, \text{cof } \mathbf{F}, J)$; Φ_θ composed with this feature map stays convex by the vector composition rule; and the remaining terms of W_{pc} are themselves convex or constant.

1. **Each feature is convex in its own block.** For a fixed unit vector $\mathbf{d}$, $\mathbf{F} \mapsto |\mathbf{F}\mathbf{d}|^2$ is a convex quadratic form in $\mathbf{F}$ (a composition of the linear map $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}$ with the convex norm-squared). For $p \in \{2, 3\}$, $x \mapsto x^p$ is convex and non-decreasing on $x \geq 0$, and $|\mathbf{F}\mathbf{d}|^2 \geq 0$; by the scalar convex-composition rule (g convex non-decreasing, f convex $\Rightarrow g \circ f$ convex), $|\mathbf{F}\mathbf{d}|^{2p}$ is convex in $\mathbf{F}$, and summing with positive weights preserves convexity, so every $Q_{k,p}^F$ is convex in $\mathbf{F}$. The identical argument with $\mathbf{F}$ replaced by $\text{cof } \mathbf{F}$ shows every $Q_{k,p}^H$ is convex in $\text{cof } \mathbf{F}$. Finally, $I_1 = |\mathbf{F}|^2$ is convex in $\mathbf{F}$, and J, J^2 are (trivially, being affine and quadratic) convex functions of the third block variable.
2. **Hence convex in the joint variable.** Each of these 15 functions, viewed as a function of the joint variable $(\mathbf{F}, \text{cof } \mathbf{F}, J) \in \mathbb{R}^{2 \times 2} \times \mathbb{R}^{2 \times 2} \times \mathbb{R}$, depends on only one block and is convex in that block, hence is convex in the joint variable (a function of a linear projection of a convex function is convex).
3. **Φ_θ composed with the feature map stays convex.** Φ_θ is convex in $\mathbf{z}_{\text{pc}}$ and non-decreasing in each coordinate (weight non-negativity at every layer, standard ICNN induction: each Softplus pre-activation is an affine, non-negative-weighted combination of convex non-decreasing quantities of the input, hence convex and non-decreasing, and Softplus itself is convex and non-decreasing, so the composition is too, by induction on layers). By the *vector* convex-composition rule (Boyd & Vandenberghe, §3.2.4: h convex and non-decreasing in each argument, each g_i convex $\Rightarrow h(g_1, \dots, g_n)$ convex), $\Phi_\theta(\mathbf{z}_{\text{pc}}(\mathbf{F}, \text{cof } \mathbf{F}, J))$ is convex in $(\mathbf{F}, \text{cof } \mathbf{F}, J)$.
4. **The remaining terms, and the sum.** The additive constant $-\Phi_\theta(\mathbf{z}_{\text{pc}}(\mathbf{I}))$ does not affect convexity, and $-r \log J$ and $\frac{\beta}{2}(J-1)^2$ are convex functions of $J > 0$ for $r, \beta > 0$. A sum of functions each convex in $(\mathbf{F}, \text{cof } \mathbf{F}, J)$ is convex in $(\mathbf{F}, \text{cof } \mathbf{F}, J)$; setting G equal to this sum proves the proposition.

Note that this establishes convexity for $(\mathbf{F}, \text{cof } \mathbf{F}, J)$ ranging over the whole product space, i.e. genuinely as three *independent* block arguments, which is exactly (22), not merely along the constrained manifold $\{(\mathbf{F}, \text{cof } \mathbf{F}, \det \mathbf{F}) : \mathbf{F} \in \text{GL}^+(2)\}$. $\square$

Remark 3. *This is a construction proof: polyconvexity is a property of the functional form (8)–(29), holding for every value of θ , not a property enforced by a training penalty that could be violated between checkpoints or degraded by future retraining with a different loss (Section 6.1). By Proposition 3 and (23), W_{pc} is automatically also rank-one convex, and this is what the numerical sample in Section 4.6.3 attempts to falsify (and fails to).*

Proposition 4 (Why $r \geq 0$ automatically). By (18), $r = 2\lambda$ with λ a non-negative combination (positive coefficients c_m , non-negative $\partial_{z_m} \Phi_\theta$) of non-negative terms, so $r \geq 0$ for any trained θ , not only the reported one.

4.6.5 A second convex core: input-convex Kolmogorov–Arnold network (ICKAN)

Every proposition above is stated for “any Φ_θ that is convex and coordinate-wise non-decreasing in $\mathbf{z}_{\text{pc}}$ ”: none of them use any property specific to the Softplus ICNN core. This subsection replaces Φ_θ with an input-convex Kolmogorov–Arnold network (ICKAN), following Thakolkaran et al. [23] and using their reference implementation (<https://github.com/mmc-group/ICKANs>), while keeping every other part of the construction (the balanced direction systems, the 15-dimensional feature vector $\mathbf{z}_{\text{pc}}$, the reference-state cancellation of Proposition 2, and the volumetric barrier of Section 4.7) identical to Section 4.6.4.

A Kolmogorov–Arnold network replaces each scalar weight of a standard layer with a learned univariate spline. In the reference implementation, every such spline is reparameterized so that its control points c_i satisfy $c_{i+1} - c_i \geq 0$ and $c_{i+2} - 2c_{i+1} + c_i \geq 0$ (a double cumulative sum of non-negative increments): the first condition makes the spline non-decreasing, the second makes it convex, for *any* value of the underlying trainable parameters. Disabling the (non-convex) residual base function and freezing the inter-layer affine maps to the identity (exactly the configuration used by Thakolkaran et al. and reused here) means every

edge of the network is simultaneously convex and non-decreasing in its scalar input, and a sum of such edges is again convex and non-decreasing by the same rule used for the ICNN. We verified this numerically before training: for a randomly initialized ICKAN core, increasing one input strictly increases the output, and the output is convex along an arbitrary line through feature space. Consequently, Propositions 2–3 apply to this model without modification, confirmed directly by measurement in Section 6.4, where the reference stress norm for the trained ICKAN model matches the ICNN model’s at machine precision.

4.7 (C6) Growth condition

Ball’s existence theorem pairs polyconvexity with a coercivity/growth condition on G , schematically

$$G(\mathbf{F}, H, \delta) \geq c_1 |\mathbf{F}|^p + c_2 |H|^q + h(\delta), \quad p, q \geq 2, \quad (30)$$

with h convex, $h(\delta) \rightarrow \infty$ as $\delta \rightarrow 0^+$ and superlinearly as $\delta \rightarrow \infty$, needed so that minimizing sequences of the total energy do not escape to infinity and a minimizer exists. Our construction supplies this structure explicitly:

- The sixth-power terms $Q_{k,3}^F, Q_{k,3}^H \sim |\mathbf{F}\mathbf{d}|^6, |\text{cof}(\mathbf{F})\mathbf{d}|^6$ give super-quadratic polynomial growth in $\mathbf{F}$ and $\text{cof } \mathbf{F}$ as any structural stretch diverges; Proposition 5 below makes this precise with an explicit, direction-independent lower bound, rather than a qualitative appeal to direction coverage.
- The volumetric term $h(J) = -r \log J + \frac{\beta}{2}(J-1)^2$ diverges as $J \rightarrow 0^+$ (impenetrability of matter: infinite energy to crush the RVE to zero volume), exactly the qualitative shape used in classical polyconvex volumetric penalties (e.g. Ciarlet–Geymonat-type terms). As $J \rightarrow \infty$, $-r \log J$ alone *decreases* without bound; it is the quadratic term that guarantees $h(J) \rightarrow \infty$ eventually, for any $\beta > 0$, since a quadratic dominates a logarithm asymptotically regardless of the constants involved. With the coefficients learned by this model ($r = 2.0390e - 01$, $\beta = 2.2357e - 09$), that crossover occurs only at $J^* \approx 9550$ (the root of $h'(J) = 0$ for $J > 1$), an area dilation never approached by this RVE’s training or test data; Figure 9 isolates $h(J)$ and confirms this directly.

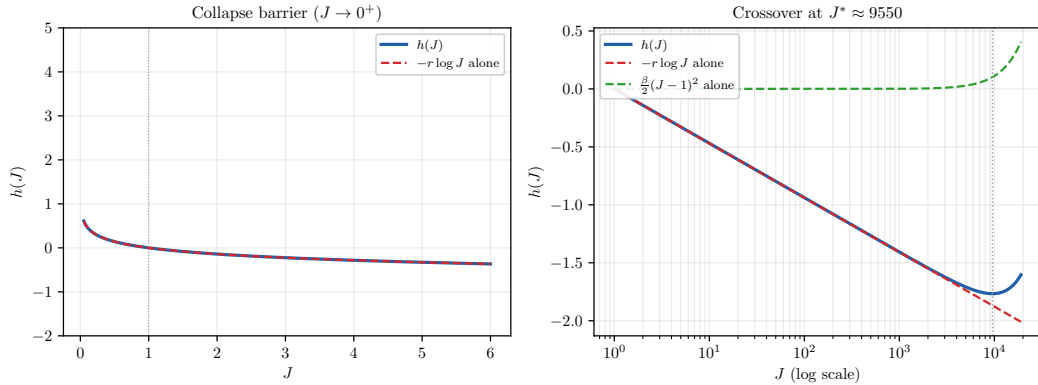


Figure 9: The volumetric barrier $h(J)$ in isolation, with no feature or network terms and no coupling to a stretching path, for the trained coefficients $r = 2.0390e - 01$, $\beta = 2.2357e - 09$. Left: near $J = 1$, $h(J)$ and $-r \log J$ alone are visually indistinguishable, and both diverge as $J \rightarrow 0^+$. Right: on a log scale, $-r \log J$ alone decreases without bound, and $h(J)$ tracks it closely until the quadratic term alone finally overtakes it near $J^* \approx 9550$, the point where $h(J)$ starts to grow. This J^* is far beyond any dilation this RVE’s data ever reaches; the total trained energy nonetheless grows steeply for realistic stretches, driven by the sixth-power terms of Proposition 5 below rather than by $h(J)$ itself (Figure 10).

Proposition 5 (A direction-independent lower bound on the sixth-power terms). *For any single balanced system $(\sum_i w_i \mathbf{d}_i \otimes \mathbf{d}_i = \mathbf{I}, w_i > 0)$ and any $\mathbf{F}$,*

$$Q_{k,3}^F = \sum_{i=1}^3 w_i (\mathbf{d}_i \cdot \mathbf{C} \mathbf{d}_i)^3 \geq \frac{1}{4} |\mathbf{F}|^6, \quad Q_{k,3}^H \geq \frac{1}{4} |\mathbf{F}|^6, \quad (31)$$

where $|\mathbf{F}|^2 = \text{tr } \mathbf{C}$ is the Frobenius norm. In particular $Q_{k,3}^F, Q_{k,3}^H \rightarrow \infty$ as $|\mathbf{F}| \rightarrow \infty$ along every direction, not merely the three structural ones: with no appeal needed to how many orientations the balanced system happens to cover.

Proof. Write $t_i = \mathbf{d}_i \cdot \mathbf{C} \mathbf{d}_i \geq 0$. Two consequences of the balance condition, both already used in the proof of Proposition 1, are all this needs: $\sum_i w_i = 2$, and, contracting the balance identity with $\mathbf{C}$,

$$\sum_i w_i \mathbf{d}_i \cdot \mathbf{C} \mathbf{d}_i = \mathbf{C} : \sum_i w_i \mathbf{d}_i \otimes \mathbf{d}_i = \mathbf{C} : \mathbf{I} = \text{tr } \mathbf{C}, \quad \text{i.e.} \quad \sum_i w_i t_i = |\mathbf{F}|^2. \quad (32)$$

Since $x \mapsto x^3$ is convex on $x \geq 0$, Jensen's inequality applied with weights $w_i/2$ (which sum to 1) gives

$$\left(\sum_i \frac{w_i}{2} t_i \right)^3 \leq \sum_i \frac{w_i}{2} t_i^3, \quad \text{i.e.} \quad \left(\frac{|\mathbf{F}|^2}{2} \right)^3 \leq \frac{1}{2} Q_{k,3}^F, \quad (33)$$

which rearranges to $Q_{k,3}^F \geq \frac{1}{4} |\mathbf{F}|^6$. The identical argument with $\mathbf{C}$ replaced by $\text{cof}(\mathbf{C})$ gives $Q_{k,3}^H \geq \frac{1}{4} |\text{cof } \mathbf{F}|^6$; in two dimensions $|\text{cof } \mathbf{F}| = |\mathbf{F}|$ exactly (the cofactor of a 2×2 matrix permutes and sign-flips entries without changing the sum of their squares), so $Q_{k,3}^H \geq \frac{1}{4} |\mathbf{F}|^6$ as well. $\square$

We do not claim to have verified the precise Sobolev-exponent matching that a fully rigorous existence proof for the associated boundary-value problem would require; that is a mathematical refinement beyond the scope of this paper. What Proposition 5 and the volumetric barrier above establish, together, is that the *qualitative* coercivity structure Ball-type theorems require is present by construction and provably direction-independent, not added as an afterthought penalty or asserted by appeal to intuition.

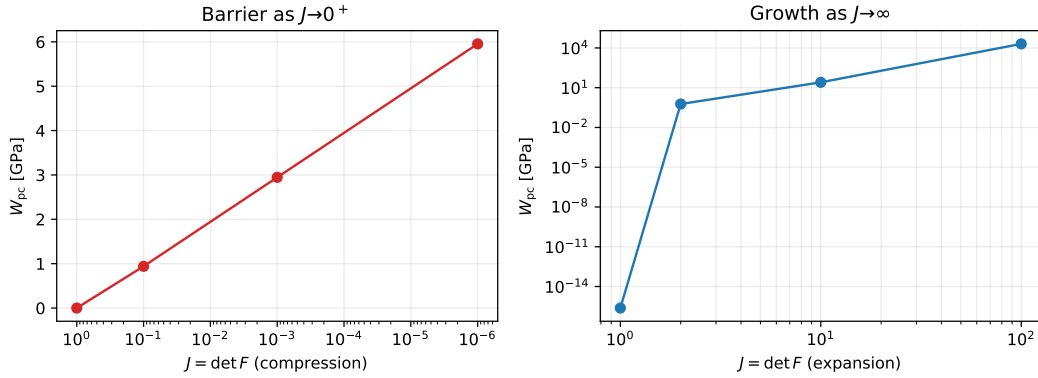


Figure 10: Total trained energy W_{pc} along the pure-dilatation path $\mathbf{F} = \text{diag}(J, 1)$: the logarithmic barrier acts as $J \rightarrow 0^+$, and the energy grows steeply as $J \rightarrow \infty$. Along this path $|\mathbf{F}|$ grows together with J , so the growth shown here for $J \gg 1$ is dominated by the sixth-power structural-stretch terms of Proposition 5, not by the volumetric term $h(J)$ in isolation; see Figure 9 for that.

Placing this against the third cluster of Section 1, which gets polyconvexity right but rarely exercises (C6) in practice. Linka & Kuhl [21], Tac et al. [22], and Thakolkaran et al. [23] all correctly use the additive, block-decoupled sufficient condition for polyconvexity (convex, non-decreasing scalar functions of I_1 , of a polyconvex modification of I_2 , and of J , summed), and Tac et al. are unusually careful to distinguish this from the strictly weaker rank-one convexity; but the growth condition (C6), while stated by all three, is uniformly *not exercised*, since Linka & Kuhl and Tac et al. both work with (nearly) incompressible materials, so $J \rightarrow 0^+$ is never actually reached by a trained model, and Thakolkaran et al.'s sole volumetric term, $K_3 = (J - 1)^2$, stays finite at $J = 0$. Our own volumetric term is exercised, not merely stated (Table 3–4 and Figures 9–10 sample W_{pc} down to $J = 10^{-6}$ for a compressible, general anisotropic material), and Section 4.6.5 shows the same construction transfers, proof-for-proof, to Thakolkaran et al.'s own ICKAN architecture: no other work surveyed here combines an ICKAN core with a general, group-free anisotropic feature set. Two further isotropic-only works round out this cluster: Tac, Sahli-Costabal and Buganza Tepole [28] originate the neural-ODE route to the same monotone-invariant sufficient condition on anisotropic (two-fixed-fiber) skin data, again without an exercised $J \rightarrow 0^+$ barrier, while Abdolazizi, Aydin, Cyron and Linka [29] apply an unconstrained Kolmogorov–Arnold network to the same isotropic invariants and decline any convexity requirement at all, with good fits and no reported instability, consistent with (not evidence against) the point Section 6.5 makes directly on our own data: losing the polyconvexity certificate does not necessarily show up as a visible failure on in-distribution data.

5 Projection-based reduced-order model

This section fixes notation for the PROM used in Section 6 as an independent, non-ANN-constitutive-law comparison point for the PANN of Section 4, applied here to the RVE and FOM of Section 2.1. The section builds a single online pipeline in three stages, not three competing constructions: a base, linear projection-based reduced basis (below); corrected by a trained nonlinear-manifold decoder, PROM-ANN (Section 5.1); evaluated with a hyper-reduced cubature rule, HPROM-ANN (Section 5.2). HPROM-ANN and D-HPROM-ANN, evaluated separately in Section 6.3, are in turn two *operating modes* of that same hyper-reduced pipeline and the same trained decoder, not two different models; HPROM-ANN runs the decoder’s prediction through the usual Newton correction of (35) to convergence, while D-HPROM-ANN simply skips that correction and uses the decoder’s own prediction directly, on the premise that it is already a sufficiently good estimate of the reduced state on its own. Section 5.1 introduces the decoder once; the split into these two modes is stated precisely where it arises, at the end of that subsection.

A standard linear PROM replaces $\mathbf{u}_f$ by its projection onto a low-dimensional subspace. Snapshots $\mathbf{u}_f(\boldsymbol{\varepsilon}_1), \dots, \mathbf{u}_f(\boldsymbol{\varepsilon}_{n_s})$ from FOM solves along the training trajectories (Fig. 3) are assembled into a snapshot matrix $\mathbf{D} \in \mathbb{R}^{n_L \times n_s}$, and a thin, truncated singular value decomposition gives an orthonormal reduced-order basis (ROB) $\mathbf{V}_{\text{tot}} \in \mathbb{R}^{n_L \times n_{\text{tot}}}$ spanning the dominant left singular subspace of $\mathbf{D}$. The displacement field is then approximated by

$$\mathbf{u}_f \approx \mathbf{u}_f^{\text{aff}}(\boldsymbol{\varepsilon}) + \mathbf{V}_{\text{tot}} \mathbf{q}_{\text{tot}}(\boldsymbol{\varepsilon}), \quad (34)$$

where $\mathbf{u}_f^{\text{aff}}(\boldsymbol{\varepsilon})$ is the affine baseline consistent with the prescribed macroscopic Dirichlet data $\mathbf{u}_d(\boldsymbol{\varepsilon})$ (so that $\mathbf{V}_{\text{tot}} \mathbf{q}_{\text{tot}}$ carries only the correction beyond that affine guess) and $\mathbf{q}_{\text{tot}} \in \mathbb{R}^{n_{\text{tot}}}$ are the generalized (POD) coordinates.

Substituting (34) into (1) and testing with the same basis (Galerkin projection, the natural choice here since the tangent operator of a hyperelastic problem is symmetric, unlike the non-self-adjoint operators that motivate a least-squares Petrov–Galerkin, LSPG, test space in the compressible-flow PROM literature [16, 38]) gives the reduced equilibrium equation

$$\mathbf{V}_{\text{tot}}^T \mathbf{r}(\mathbf{u}_f^{\text{aff}}(\boldsymbol{\varepsilon}) + \mathbf{V}_{\text{tot}} \mathbf{q}_{\text{tot}}; \boldsymbol{\varepsilon}) = \mathbf{0}, \quad (35)$$

a system of $n_{\text{tot}} \ll n_L$ nonlinear equations, again solved by Newton–Raphson. As documented for this exact class of problem (propagating features, evolving nonlinear regions) by Barnett and Farhat [13] and by Hernández’s own metamaterial and damage examples [19], a single linear subspace of dimension small enough to be worth reducing to generally cannot compress a nonlinear problem’s solution manifold efficiently (the Kolmogorov n -width barrier), which restricts the efficiency of linear-subspace reduction here too, since achieving good accuracy can require keeping the retained dimension itself large. The nonlinear-manifold construction of this paper, PROM-ANN, addresses exactly this point.

5.1 Nonlinear-manifold PROM: PROM-ANN

Following the general nonlinear-PMOR strategy of decomposing the ROB into a small *primary* (retained) part and a larger *secondary* (discarded) part whose amplitude is reconstructed by a trained nonlinear map, introduced for fluid-dynamics PMOR as the PROM-ANN/HPROM-ANN framework by Barnett, Farhat and Maday [16] and generalized to interpretable Gaussian-process and radial-basis-function closures by Ares De Parga et al. [38] for settings with fewer affordable training snapshots. This project’s own RVE is cheap enough to sample densely, so the original neural-network regressor is used throughout instead. PROM-ANN denotes this paper’s own instance of that framework, the same base PROM above, still using every Gauss point of the original mesh, but with the discarded modes’ amplitude reconstructed by a trained decoder instead of being dropped outright. The total basis is partitioned as $\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \bar{\mathbf{V}}]$, $\mathbf{V} \in \mathbb{R}^{n_L \times n}$, $\bar{\mathbf{V}} \in \mathbb{R}^{n_L \times \bar{n}}$, $n \ll \bar{n}$, and the displacement field is approximated as

$$\mathbf{u}_f \approx \mathbf{u}_f^{\text{aff}}(\boldsymbol{\varepsilon}) + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q}), \quad (36)$$

with $\mathbf{q} \in \mathbb{R}^n$ the primary (retained) generalized coordinates and $\mathcal{N} : \mathbb{R}^n \rightarrow \mathbb{R}^{\bar{n}}$ a trained neural network that reconstructs the secondary (discarded) amplitudes ($n = 3$, $\bar{n} = 36$ for the trained model used throughout this paper).

It is natural to ask why the primary coordinates are not simply the first three raw POD coordinates. The raw POD coordinates $\mathbf{Q}_{\text{tot}} = \mathbf{V}_{\text{tot}}^T \mathbf{S}$ of a snapshot matrix $\mathbf{S}$ are optimal, by construction, for linear reconstruction accuracy, since no other rank- n subspace captures more snapshot energy. They are not, however, optimized for what the decoder in (36) actually needs downstream: a coordinate that a regression map $\boldsymbol{\varepsilon} \mapsto \mathbf{q}$ can learn accurately. Figure 11 shows why this matters concretely for this RVE: the first three raw POD coordinates of the training trajectories (Fig. 3) trace out a curved, self-intersecting ribbon in $\mathbb{R}^3$

(left panel) that bears little resemblance to the simple, boustrophedon sweep of the macro-strain domain itself (right panel): a nonlinear, non-monotonic relationship that is unnecessarily hard for a regression network to learn well. We therefore do not use $\mathbf{Q}_{\text{tot}}$ directly; instead, $\mathbf{V}$ and $\mathbf{q}$ are identified via the same least-squares primary/secondary basis-identification procedure used by Hernández’s MAW–ECM paper [19] for a different homogenization RVE: a linear transformation of the raw POD coordinates, fitted so that the resulting *structured* coordinates track the macro-strain path as closely as possible in a least-squares sense, while remaining an exact orthonormal change of basis within the primary subspace (so (36)’s reconstruction accuracy is unaffected). The middle panel of Figure 11 shows the effect on this RVE’s own data. The same ten training trajectories, expressed in the structured coordinates, recover the simple, near-affine sweep pattern of the macro-strain domain itself, at a verified affine-fit accuracy of 6.9×10^{-4} relative L^2 error ($R^2 \geq 0.99999$ componentwise) and a primary/secondary orthogonality residual of 2.3×10^{-15} (machine precision). This is exactly what makes the direct macro-strain-to-coordinate regression in (36) tractable, because $\mathcal{N}$ only has to learn the much smaller residual nonlinearity between this already near-affine structured coordinate and the discarded secondary amplitudes, not the raw POD manifold’s own curvature.

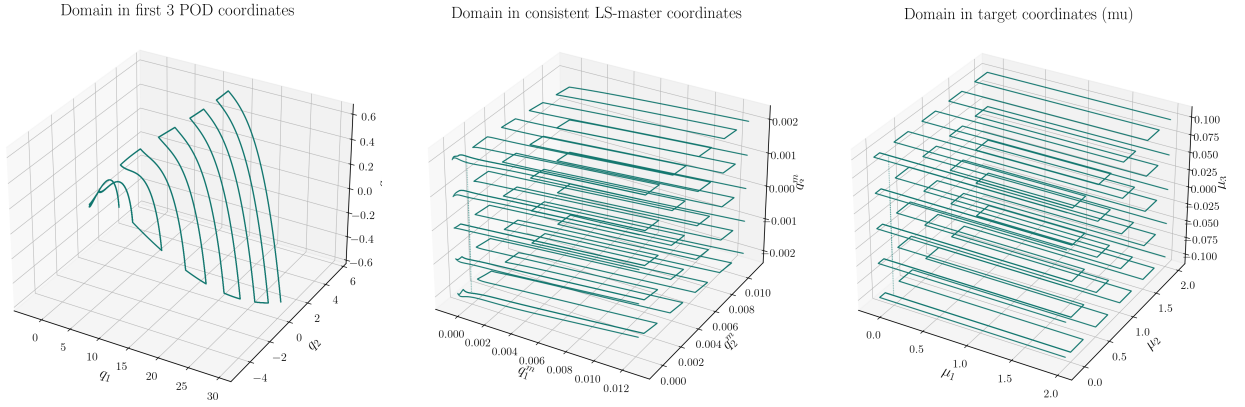


Figure 11: The training domain in three coordinate systems, all built from the same 109 460 training-trajectory snapshots of this RVE. Left: the first three raw POD coordinates $(q_{\text{tot},1}, q_{\text{tot},2}, q_{\text{tot},3})$, a curved, self-intersecting ribbon that is POD-optimal for reconstruction but not well suited to regression. Middle: the same snapshots in the structured, least-squares-fitted primary coordinates $\mathbf{q}$ actually used as (36)’s input, recovering a near-affine, boustrophedon sweep. Right: the macro-strain domain itself, $\boldsymbol{\varepsilon} = (E_{11}, E_{22}, \gamma_{12})$, for comparison. The middle and right panels are visually close by construction: that closeness is exactly the least-squares fit being verified, not assumed.

The decoder network $\mathcal{N}$ itself (Figure 12) is a feedforward network with four hidden layers of 128 units each and SiLU activations, trained to map the 3-dimensional structured coordinate directly to the 36-dimensional secondary amplitude.

In this project, $\mathbf{V}$ and $\mathbf{q}$ are identified with the macro-strain $\boldsymbol{\varepsilon}$ itself, an *input-driven* manifold, valid here because the problem is static and hyperelastic (no history dependence), so the equilibrium state is uniquely parameterized by $\boldsymbol{\varepsilon}$. Consequently $\mathcal{N}$ is trained once, offline, on the training trajectories to map macro-strain directly to the reconstructed secondary coordinates, and no reduced equilibrium equation of the form (35) needs to be solved online at all if $\mathcal{N}$ ’s prediction is trusted directly; alternatively, (36) can still be substituted into a Galerkin-projected residual and corrected by Newton iterations, exactly as in (35). These are precisely the two HPROM–ANN operating modes evaluated in Section 6.3: the *iterative* mode (Newton correction to convergence) and the *direct* mode, D-HPROM–ANN (zero Newton iterations, $\mathcal{N}$ ’s output used as-is).

5.2 Hyperreduction: HPROM-ANN via the manifold-adaptive-weight empirical cubature method (MAW–ECM)

HPROM-ANN is PROM-ANN above with one further reduction: replacing the full Gauss-point loop itself by a sparse, learned cubature rule. Even after (36), evaluating the residual (1) (or the reduced residual (35)) still requires a full loop over all $n_{\text{gp}} = \sum_e n_{\text{gp}}^{(e)}$ Gauss points of the original FOM mesh, since the stress $\mathbf{S}_{e,g}$ must be evaluated from the true microscopic constitutive law at every one of them. Reindexing the element/Gauss-point pairs (e, g) of (1) by a single global index $g \in \{1, \dots, n_{\text{gp}}\}$ (as is standard in the ECM literature), so that $\mathbf{r}_g := \mathbf{L}_e^T \mathbf{B}_{e,g}^T \mathbf{S}_{e,g}$ and $\mathbf{S}_g := \mathbf{S}_{e,g}$ denote the residual and stress contribution of global Gauss point g , hyperreduction replaces the full Gauss-point sum by a sparse cubature rule over a small subset

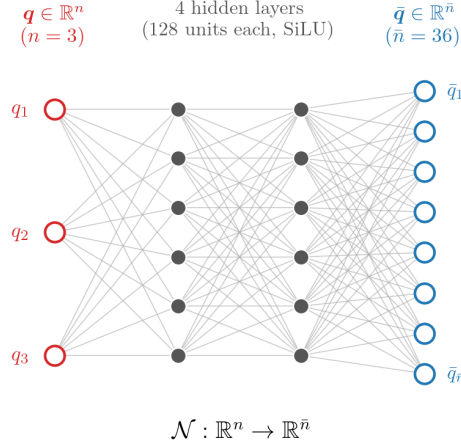


Figure 12: The decoder network $\mathcal{N} : \mathbb{R}^n \rightarrow \mathbb{R}^{\bar{n}}$ of (36) for this RVE's trained HPRM-ANN model ($n = 3$, $\bar{n} = 36$); only a representative subset of the 128 units per hidden layer is drawn.

$\mathcal{Z} \subset \{1, \dots, n_{\text{gp}}\}$ of retained points with positive weights $\{\omega_g\}_{g \in \mathcal{Z}}$:

$$\sum_{g \in \mathcal{Z}} (\mathbf{V}^T \mathbf{r}_g)(\mathbf{u}_f; \boldsymbol{\varepsilon}) \omega_g = \mathbf{0}, \quad \mathbf{S}_{\text{macro}}(\boldsymbol{\varepsilon}) \approx \frac{1}{|\Omega_0|} \sum_{g \in \mathcal{Z}_\sigma} \mathbf{S}_g(\mathbf{u}_f; \boldsymbol{\varepsilon}) \omega_{\sigma, g}, \quad (37)$$

one rule ($\mathcal{Z}$, weights ω_g) tailored to integrate the projected *equilibrium* residual accurately, and, in general, a second, independently constructed rule ($\mathcal{Z}_\sigma$, weights $\omega_{\sigma, g}$) tailored to the homogenized-*stress* output functional: the two point sets and weight sets are not required to coincide, and Fig. 13 shows both retained element sets for this RVE side by side.

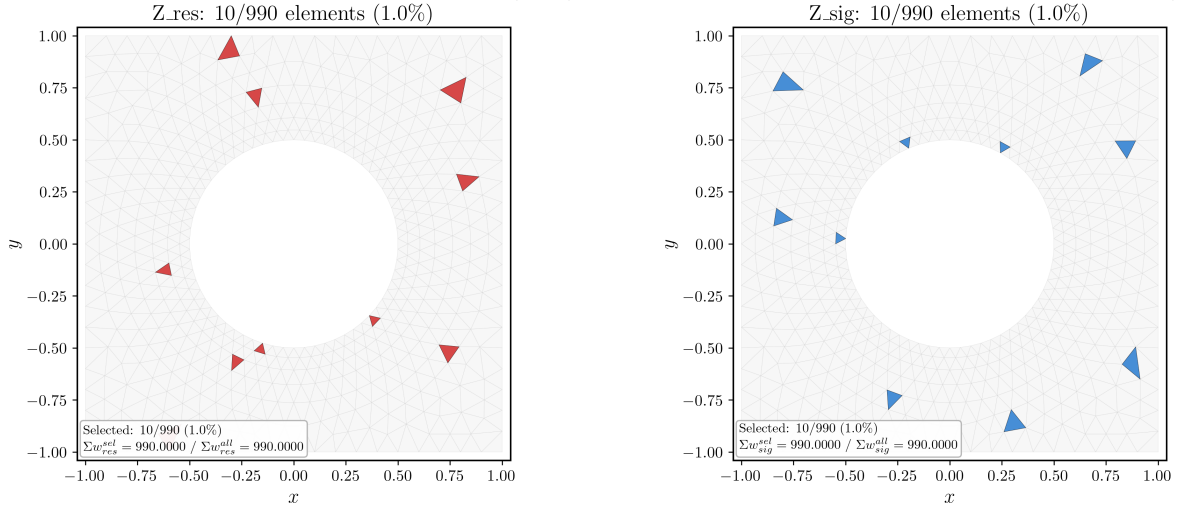


Figure 13: MAW-ECM hyperreduced integration supports for this RVE's canonical manifold-adaptive-weight configuration. Left: the $|\mathcal{Z}| = 10$ Gauss points retained for the projected equilibrium residual. Right: the $|\mathcal{Z}_\sigma| = 10$ Gauss points retained for the homogenized-stress output functional. The two supports are constructed independently and, as visible here, are not required to select the same elements.

The base weights, for a single fixed macro-strain state, are found offline by the ECM of Hernández, Caicedo and Ferrer [18], using the greedy active-set sparse non-negative least-squares algorithm of Hernández's own later formulation [11]: starting from an empty point set, the element whose residual-basis column best explains the current integration-error vector is added at each step, the weights on the active set are recomputed by a constrained least-squares solve, and any element whose weight has gone negative is purged and re-solved, until the truncated-SVD basis of the training-snapshot integrand space is integrated to within a prescribed

tolerance. This is in the same spirit as (and predates) the energy-conserving-sampling-and-weighting (ECSW) family used for Petrov–Galerkin hyperreduction in the fluids PMOR literature [39, 40, 41] and the empirical quadrature procedure [42, 43, 44]. A model that uses a single such *fixed* weight set for every macro-strain state is an ECM-hyperreduced HPROM.

The manifold-adaptive-weight idea behind MAW–ECM, introduced by Hernández [19], instead lets ω_g (and $\omega_{\sigma,g}$) vary smoothly with the reduced state, here the macro-strain, so that the same support set $\mathcal{Z}$ stays accurate over a wider region of the strain path than a fixed rule could. Hernández’s own construction achieves this smoothness through a graph-Laplacian-regularized pruning-and-regression scheme over a discretized manifold of candidate states, and the construction deployed for this RVE follows the same two-stage idea, in the same spirit, though not a line-for-line reproduction of the published implementation.

Stage one: graph-regularized support pruning. Starting from a larger candidate support already produced by the fixed-weight ECM procedure above (one candidate weight vector per training state), a two-phase backward-elimination procedure prunes this candidate set down to the final, fixed support $\mathcal{Z}$ (or $\mathcal{Z}_\sigma$) reported in Fig. 13. In phase one, candidates are removed one at a time, independently at each training state, whenever the remaining active set can still satisfy that state’s linear moment-matching constraint by a least-squares reweighting with no weight going negative, until an intermediate support size of 40 points is reached. In phase two, elimination continues jointly across states: a k -nearest-neighbor graph ($k = 8$) is built over a uniform subsample of 500 training states (over the reduced coordinate $\mathbf{q}$ for the equilibrium-residual rule, or over the macro-strain $\boldsymbol{\varepsilon}$ itself for the two homogenized-stress rules), and, at each step, every remaining low-activity candidate is trial-removed by solving a constrained quadratic program that keeps every sampled state’s weight vector close to its previous value while penalizing the graph-Laplacian roughness of the weight field across neighboring states (smoothing weight $\alpha = 10^2$); the removal that leaves the smoothest resulting weight field is kept, subject to the same non-negativity and moment-matching constraints as phase one, until the final target size of 10 points is reached. This graph-regularized second phase is what actually supplies the manifold-adaptive smoothness Hernández’s construction is designed to give; it is a genuine, non-trivial regularization step acting jointly across training states, not a relabeling of independent per-state ECM solves.

Stage two: continuous weight-field regression. Phase two’s output is the final fixed support together with a table of graph-smoothed weight vectors, one per sampled training state: still a discrete object that must be evaluated at arbitrary, unseen states online. This table is fit by one of two regressors depending on the rule: a Gaussian radial-basis-function model with a polynomial trend for the equilibrium-residual rule, and a softmax-output neural network for the two homogenized-stress rules (non-negativity and the fixed row sum $\sum_g \omega_g$ guaranteed by the softmax itself), trained with a loss combining a Kullback–Leibler term and a mean-squared-error term against the pruned weight table, plus a physics-exactness penalty that enforces the linear integration constraint over the *full* training set, not only the 500-state subsample used for pruning. This follows Hernández’s manifold-adaptive-weight idea in spirit rather than reproducing the published algorithm line for line. The offline scripts that produced the specific support and weights deployed for this RVE are not part of this project’s own contribution; the trained weights themselves are, and are exactly what every HPROM–ANN number in this paper is computed from.

More broadly, related work on reduced-order modeling for computational homogenization [10, 11, 45] traces back at least to the POD-based reduced model multiscale method (R3M) for finite-strain hyperelastic homogenization of Yvonnet and He [6], with subsequent hyperreduced and multilevel PROM treatments of nonlinear multiscale structural models by Zahr, Avery and Farhat [7], He, Avery and Farhat [8], and Fritzen and Hodapp [9], and, from a purely data-driven angle, recent deep-learning-based multiscale surrogates for inelastic (crystal plasticity) and heterogeneous-material homogenization problems by Liu et al. [46] and Aldakheel et al. [47]. Related work on hyperreduction for Petrov–Galerkin ROMs and parallel/digital-twin ROM workflows by the same author group [48, 49], and on machine-learning-assisted homogenization of masonry by two of this project’s collaborators [50], similarly informs this project without being part of its own contribution.

Figure 14 puts these pieces together, collecting the full online HPROM–ANN/D-HPROM–ANN pipeline of this section into a single diagram: the macro-strain input and its affine boundary data (Section 2.1), the structured primary coordinate that initializes the state (Section 5.1), the two operating modes (the Newton-corrected, ECM-sparse Galerkin residual of (35)/(37) for HPROM–ANN, or the direct, uncorrected decoder evaluation for D-HPROM–ANN), the shared decoder state map of (36), and the hyper-reduced homogenized-stress output of (37).

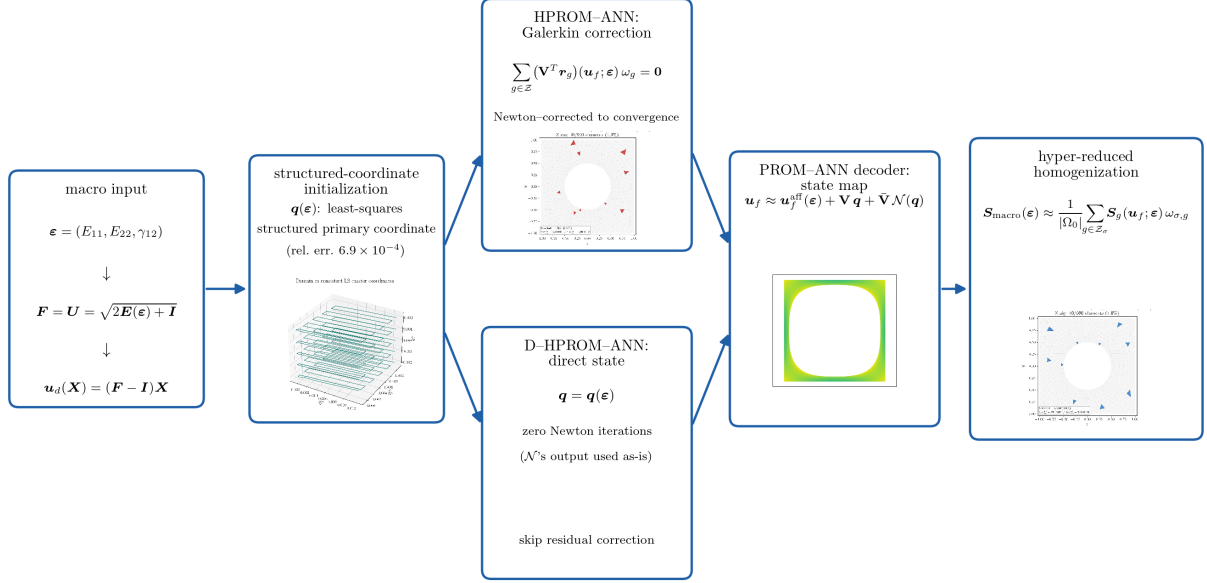


Figure 14: The full online HPROM-ANN/D-HPROM-ANN workflow for this RVE, from a prescribed macro-strain ϵ to the hyper-reduced homogenized stress $\mathbf{S}_{\text{macro}}(\epsilon)$. The two middle boxes are the two operating modes evaluated in Section 6.3: HPROM-ANN (top, Newton-corrected) and D-HPROM-ANN (bottom, direct, zero Newton iterations); both feed the same trained decoder (36) and the same hyper-reduced stress rule. Mesh thumbnails: the same $\mathcal{Z}/\mathcal{Z}_\sigma$ supports of Fig. 13 and the same displacement field of Fig. 5.

6 Numerical experiments and results

6.1 Loss design: an equivalent-stress-oriented training choice

Sections 4.2–4.7 are properties of the energy functional (8) and hold regardless of how the model is trained. The training loss is a different kind of choice: a purely statistical one, made because it measurably improves accuracy on the quantity of engineering interest, and it does not touch (8) at all. It is worth describing carefully, because none of the polyconvex-PANN constructions of Table 1 discuss a risk we found here. Per-component loss normalization, used in its most extreme form by As’ad et al. [3], can measurably hurt accuracy on the physically relevant equivalent stress when the material’s stress components differ by orders of magnitude.

Every model reported in this paper is trained on the plain two-term sum

$$\mathcal{L} = 0.65 \mathcal{L}_W + \mathcal{L}_{S,\text{agg}}, \quad (38)$$

where $\mathcal{L}_W$ normalizes the mean squared energy error by its mean squared target, and $\mathcal{L}_{S,\text{agg}}$ normalizes the mean squared error of the full stress vector (S_{11}, S_{22}, S_{12}) by the mean squared norm of the full target vector, so it is dominated by whichever components carry the most physical stress magnitude. In both terms the predicted stress is $\partial W_{\text{pc}}/\partial \epsilon$ of the predicted energy, as in (4).

We arrived at (38) by rejecting an alternative that looked reasonable at first. In this RVE, S_{12} is roughly two orders of magnitude smaller than S_{11}, S_{22} along both the training trajectories and the test trajectory alike, which makes it tempting to give it its own dedicated weight. The previously certified version of this model did exactly that, blending in a second, componentwise stress term,

$$\mathcal{L}_S(\gamma) = (1 - \gamma) \mathcal{L}_{S,\text{agg}} + \gamma \mathcal{L}_{S,\text{comp}}, \quad \gamma \in [0, 1], \quad (39)$$

where $\mathcal{L}_{S,\text{comp}}$ is the equally weighted mean of the three componentwise relative-squared errors (each of S_{11}, S_{22}, S_{12} normalized by its own mean squared target before averaging, rather than by the norm of the full vector). At $\gamma = 0.65$, the value used previously, the optimizer is forced to reduce S_{12} ’s relative error to the same standard as S_{11}, S_{22} , even though S_{12} contributes negligibly to the plane-stress equivalent (von Mises) stress

$$\sigma_{\text{eq}} = \sqrt{S_{11}^2 - S_{11}S_{22} + S_{22}^2 + 3S_{12}^2}, \quad (40)$$

the quantity of actual engineering interest here. As’ad et al. [3] push exactly this kind of per-component weighting to its most extreme form.

Table 5 isolates this choice from the other change made at the same time, widening the ICNN. Widening alone, with $\gamma = 0.65$ unchanged, gives a small improvement over the original network; setting $\gamma = 0$ instead, dropping $\mathcal{L}_{S,\text{comp}}$ entirely, gives essentially all of the remaining gain, and, somewhat counter-intuitively, improves S_{12} ’s own componentwise error rather than trading it away for a better fit elsewhere. The likely explanation is that the componentwise term was forcing the optimizer to chase a noisy, two-orders-of-magnitude-smaller relative target at the expense of the rest of the fit, rather than any genuine tension between the components. Since $\gamma = 0$ is what actually works best, (39) appears here only as the rejected alternative that motivates (38), and plays no further role in this paper.

Table 5: Ablation of the two changes, test-trajectory relative L^2 error. Same seed and optimizer settings throughout; only the ICNN width and the componentwise-blend weight γ of (39) differ.

Configuration	W	S	σ_{eq}
Original (32, 32, comp. weight 0.65)	1.848%	2.643%	2.172%
Width only (64, 64, 32, comp. weight 0.65)	1.641%	2.359%	1.986%
Final (64, 64, 32, comp. weight 0.00)	0.420%	0.550%	0.507%

6.2 Training protocol

The model uses all 109 460 saved states of the ten training trajectories described in Section 2.2, an ICNN of widths (64, 64, 32), and the loss weights (0.65, 1.00) of (38) (i.e. $\gamma = 0$ in the rejected componentwise blend (39) of Section 6.1). It is trained in CPU float64 because its energy–stress loss differentiates the energy twice; its best epoch by the tracked training score is 4620. The test trajectory is not used, in any form, during training.

6.3 Test-trajectory assessment: all six model variants

Table 6 reports relative L^2 errors against the direct FOM labels, including the plane-stress equivalent (von Mises) stress (40), for all six model variants considered in this paper: the four ANNs of Section 4.2.1 (tiers 1–3b), and the two hyper-reduced projection-based ROMs used elsewhere in this project as an independent, non-ANN-constitutive-law point of comparison on the identical test path. HPRM–ANN and D-HPRM–ANN are two operating modes of the same hyper-reduced solver and the same trained solution-manifold ANN (mapping macro-strain directly to reduced POD coordinates via the primary/secondary decoder of Section 5.1, unrelated to the constitutive-law ANNs of tiers 1–3b), rather than two different models. HPRM–ANN runs the full ECM-weighted Newton corrector to convergence at every step (up to 25 iterations), while D-HPRM–ANN sets the corrector iteration count to zero and simply evaluates the manifold ANN’s direct macro-strain-to-reduced-state prediction, with no equilibrium solve at all. Both reconstruct stress from the true microscale Neo-Hookean law at ECM-selected Gauss points rather than from a learned constitutive surrogate, so they measure a different source of error, hyper-reduction and manifold-regression error, than tiers 1–3b’s constitutive-surrogate error. (This construction is not itself a contribution of this paper; Section 5 fixes its notation, cites its sources in full, and places it against the broader reduced-order-modeling literature.)

There is a further complication worth spelling out. HPRM–ANN’s native output is a naive volume-average of the converged microscopic strain/stress field, not the exactly energy-conjugate macro-stress that tiers 1–3b are trained and evaluated against (the same regression-vs.-conjugate distinction of Section 4.2.1, one level up the pipeline). For a mixed Dirichlet/free boundary-value problem these two macroscopic stress definitions need not agree, and empirically they do not; a direct check found the two differing by roughly 30% on this RVE. Comparing HPRM–ANN’s raw output against the other five rows below would therefore not be comparing like with like, so instead we compute the same energy-conjugate quantity for HPRM–ANN’s own converged state. By the envelope theorem at an equilibrium point, this reduces to a reaction-force-weighted sum over the Dirichlet-constrained boundary nodes, with no re-solving or finite differences needed. We verified that this reaction-force procedure reproduces the training data’s own energy-conjugate stress to a relative L^2 error of $5.90e - 10$, effectively machine precision, on a held-out subset of the training trajectories, before applying it to the test-trajectory HPRM–ANN/D-HPRM–ANN states reported below.

Tier 1 has no energy column, since Section 4.2.1 shows no potential exists in that model, and HPRM–ANN and D-HPRM–ANN have none either, because the reaction-force procedure above recovers the energy-conjugate stress directly from boundary reactions, and extending it to a comparable normalized energy is a

Table 6: Test-trajectory relative L^2 errors against direct FOM labels, all six model variants.

Model	W	S	S_{11}	S_{22}	S_{12}	σ_{eq}
Pure regression (tier 1)	–	0.054%	0.048%	0.048%	9.862%	0.033%
Free hyperelastic ANN (tier 2)	0.189%	0.790%	0.886%	0.659%	44.698%	0.488%
Polyconvex ICNN (tier 3a)	0.420%	0.550%	0.591%	0.504%	3.152%	0.507%
Polyconvex ICKAN (tier 3b)	0.373%	0.554%	0.647%	0.439%	4.282%	0.538%
HPROM-ANN	–	0.174%	0.156%	0.164%	26.631%	0.131%
D-HPROM-ANN	–	0.202%	0.203%	0.202%	1.554%	0.166%

natural refinement we did not attempt here. Tier 2’s componentwise S_{11} , S_{22} errors are somewhat elevated relative to the two certified cores (0.6–0.9% vs. 0.4–0.65%) but remain sub-percent; the qualitatively different result is its S_{12} error, more than an order of magnitude worse than either certified core, and its von Mises accuracy relative to the certified tiers, discussed in Section 6.5.

The full six-model test-trajectory comparison (energy for the three energy-capable tiers, and every stress component and the von Mises equivalent for all six model variants of Table 6) is given later in Figure 15, once Section 6.5 has introduced the sixth (uncertified) variant; the corresponding relative- L^2 -error bars are in Figure 16.

Figure 10 above already shows this trained response along the same pure-dilatation path; no FOM reference is shown there since it is a model-intrinsic property, not a fit to data. The learned volumetric coefficients are $r = 2.0390e - 01$ and $\beta = 2.2357e - 09$ in normalized energy units. A separate broad positive-definite sampling audit found a minimum energy of $5.898e - 04$ GPa; this is a useful physical diagnostic, not a proof of global non-negativity, and is independent of the (C1)–(C6) guarantees above, all of which hold regardless of its value.

6.4 A second convex core: ICKAN results

Section 4.6.5 shows that the same construction of Section 4.6.4, the balanced direction systems, the 15-dimensional feature vector, the reference-state cancellation, and the volumetric barrier, carries over unchanged to the ICKAN core following Thakolkaran et al. [23], and explains why the same propositions apply to it without modification (Section 4.7 places this relative to the ICKAN literature). Measurement confirms it directly. The reference stress norm for the trained ICKAN model is $4.049e - 24$, at machine precision, exactly as for the ICNN model.

The ICKAN core used here has hidden widths (8, 8), a grid of 6 intervals, and spline order 3. It is trained with the same loss (38) as the ICNN model, reaching its best tracked training score at epoch 1100; the ICNN and ICKAN rows of Table 6 and Figure 15 report its test-trajectory accuracy alongside the ICNN model. The smallest sampled rank-one curvature is $3.479e - 02$ GPa, and the broad sampled-energy audit gives a minimum of $5.834e - 04$ GPa: both positive, as for the ICNN model, consistent with the polyconvexity certificate without amounting to a proof of it.

The two cores turn out to be close, with neither uniformly better. The ICKAN reaches comparable accuracy with a hidden-layer parameter count roughly two orders of magnitude smaller than the ICNN’s (64, 64, 32), though each epoch costs more because of the spline evaluations. Both certificates are architectural and independent of which core is used, so choosing between them in practice comes down to parameter budget and training cost rather than admissibility.

6.5 What breaks without the certificate: a non-polyconvex baseline

As’ad et al. [3] and Klein et al. [1] each include a flexible, uncertified baseline in their own work for exactly this reason (Section 4.6.2): to show what the certificate is protecting against. As’ad et al. show Newton’s method diverging under load; Klein et al. show their flexible model losing ellipticity even for small deformations (their Fig. 5). We repeat that exercise on this project’s own RVE, using the flexible “free” anisotropic PANN already introduced for the original, three-model version of this study, an unconstrained Softplus MLP on the material components of $\mathbf{C}$ with no convexity constraint anywhere in the architecture, retrained with the identical loss (38) used for the two certified models above so that the comparison isolates the convexity architecture as the only difference.

Trained this way, the free model reaches 0.189% energy error and 0.488% von Mises error on the test trajectory, comparable to, and on this metric marginally better than, the certified polyconvex models of Table 6 (its S_{12} error, 44.698%, is much worse, consistent with Section 6.1). Figure 15 shows its von Mises trajectory is visually indistinguishable from the certified models and from the direct FOM. Accuracy alone,

in other words, gives no hint that a polyconvexity certificate is missing. Abdolazizi et al. made the same observation about their uncertified CKAN [29] on the basis of their own reported results (Section 4.7); here it is demonstrated directly on our own model and our own data.

The absence of the certificate is nonetheless real and findable, once the right test is applied. We sampled 2000 random rank-one perturbations $\mathbf{F} + t \mathbf{a} \otimes \mathbf{b}$ (unit $\mathbf{a}, \mathbf{b}$, log-stretch range ± 1 , i.e. stretches in $[e^{-1}, e^1]$) and evaluated $d^2/dt^2 W$ at $t = 0$ in float64, the same falsification attempt of (25) run on the certified models in Section 4.6.3, at a larger sample count. Every sample was non-negative for the certified ICNN and ICKAN models (Table 6’s discussion and Section 6.3); for the free model, 148 of 2000 samples (7.400%) are negative, with a minimum of $-1.001e + 00$ GPa, comparable in magnitude to the certified models’ minimum *positive* sampled curvature. One example violation is

$$\mathbf{F} = \begin{bmatrix} 0.534 & -0.372 \\ -0.387 & 0.728 \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} 0.939 \\ -0.345 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -0.064 \\ -0.998 \end{bmatrix}, \quad \left. \frac{d^2 W}{dt^2} \right|_{t=0} < 0, \quad (41)$$

a moderate, unremarkable compressive-shear state ($\det \mathbf{F} \approx 0.24$), not an extreme or contrived one. This is exactly what Section 4.6.3 argues. Rank-one convexity is the correct necessary consequence of polyconvexity, and without an architectural guarantee it can fail at ordinary deformation states while every accuracy metric looks fine. The certified models in this paper do not merely happen to avoid this on the states sampled; Proposition 3 proves they cannot fail it anywhere, by construction.

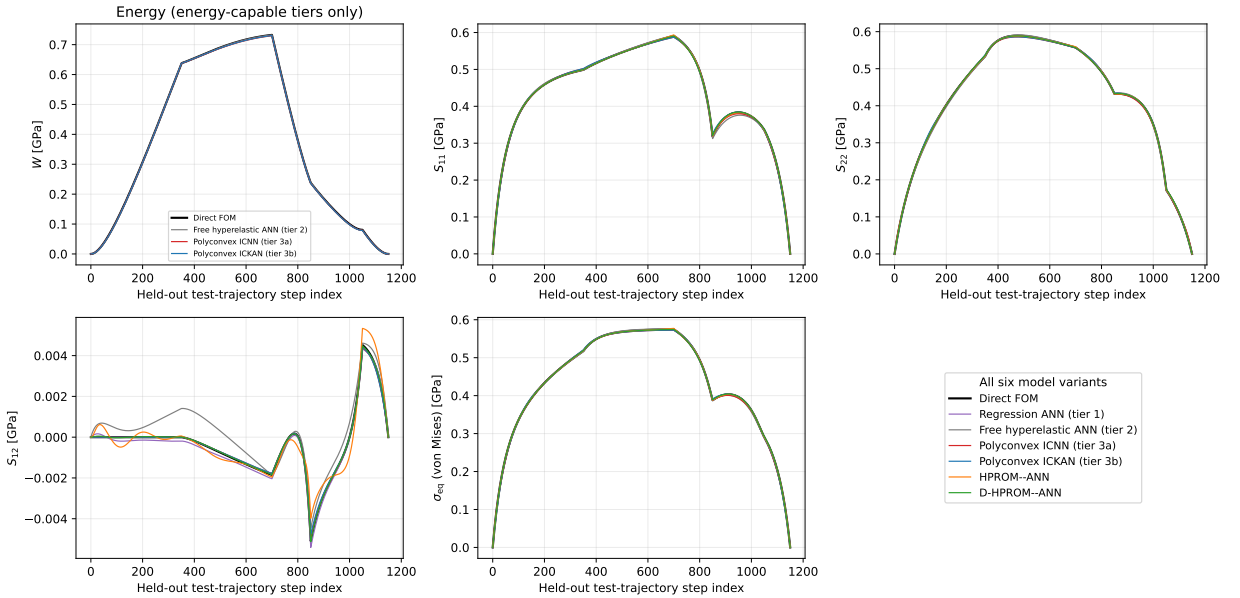


Figure 15: Test-trajectory comparison, direct FOM vs. all six model variants of Table 6 (the tier-1 pure-regression ANN, the tier-2 free hyperelastic ANN, the two certified polyconvex cores, and both HPROM-ANN operating modes): the three second Piola-Kirchhoff components S_{11} , S_{22} , S_{12} and the von Mises equivalent σ_{eq} , each shown for all six variants, plus energy W , shown only for the three energy-capable tiers (tier 2, tier 3a, tier 3b), since the tier-1 pure-regression model and both HPROM-ANN modes recover stress directly, without an underlying scalar potential, so no energy curve exists for them to plot (Section 4.2.1 for tier 1; Section 6.3’s discussion of the reaction-force procedure for the two HPROM-ANN modes). Every plotted variant is visually indistinguishable from the FOM and from each other on this test path (including the free model, whose lack of a polyconvexity certificate is not visible here), which is exactly why Sections 4.2.1 and 6.5 audit them directly (cyclic work, rank-one curvature) rather than relying on this figure.

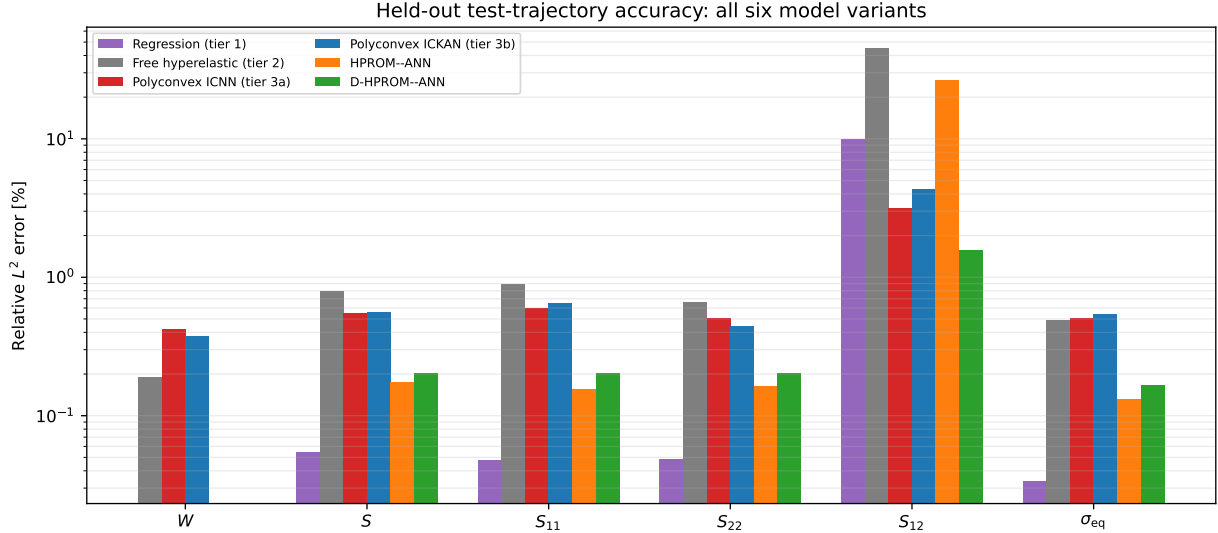


Figure 16: Test-trajectory relative L^2 error, all six reported quantities of Table 6, for all six model variants: the tier-1 pure-regression ANN, the tier-2 free hyperelastic ANN, the two certified polyconvex cores, and both HPROM-ANN operating modes. The uncertified ANNs (tiers 1–2) are competitive with, and on several components better than, the certified models on every accuracy metric shown here: consistent with Sections 4.2.1 and 6.5’s point that accuracy alone cannot distinguish a certified model from an uncertified one; only the cyclic-work and rank-one curvature audits can.

6.6 A genuine FE^2 demonstration: Cook’s membrane

Every result so far certifies the energy functional itself, or evaluates it at the level of the RVE’s own test trajectory. As’ad et al. [3] and Klein et al. [1] go one step further and show that an uncertified model can make Newton’s method fail inside an actual structural solve; this section does that too, but also goes a step further still. At every macroscopic Gauss point of a structural solve, the macroscopic Newton loop needs a homogenized stress and tangent from “the microscale.” This section supplies that need three distinct ways, all deployed on the identical macroscopic problem so they can be compared directly against one another, and against the true multiscale answer, rather than only against each other’s own test trajectory: (i) each of the four PANN tiers (regression, tier 1; free hyperelastic, tier 2; polyconvex ICNN, tier 3a; polyconvex ICKAN, tier 3b), evaluated directly – a fully-amortized, trained surrogate for the *entire* microscale solve, effectively free online, since every bit of microscale physics it uses was compressed into its trained weights offline; (ii) D-HPROM-ANN and HPROM-ANN (Section 5), nested as the actual RVE-level solver at every Gauss point – a genuine, if reduced-order, online microscale solve, not a stand-in, denoted D-HPROM-ANN- FE^2 and HPROM-ANN- FE^2 below to distinguish this nested deployment from the same trained models’ test-trajectory evaluation of Table 6; and (iii) the true, non-reduced 990-element RVE FOM itself, nested the same way, with no reduction of any kind, denoted FOM- FE^2 – the actual multiscale ground truth this whole hierarchy approximates. To our knowledge this is the first time a physics-augmented constitutive surrogate, a reduced-order projection-based surrogate, and a genuinely nested full-order microscale solve have all been compared directly, on the identical macroscopic problem, rather than only against each other on a test trajectory.

The problem is Cook’s membrane: a tapered panel with corners $(0, 0)$, $(48, 44)$, $(48, 60)$, $(0, 44)$ (SI units), clamped on its left edge and loaded by a ramped, in-plane shear force on its right edge, a standard large-deformation benchmark in the nonlinear finite-element literature, not tied to any single surveyed paper. It has no hole and no periodic structure, and its dominant loading is tensile, matching, rather than fighting, this project’s own training range (Section 2.2): the compressive and shear excursions this panel actually develops stay inside, or close to, the training window, so the comparison below is not a test of extrapolation. The final load (2.08×10^8 N, ramped linearly over 20 steps) was chosen precisely because it is the largest load, on this mesh, for which the certified models’ own strain state ($E_{11} \in [-0.039, 0.012]$, $\gamma_{12} \in [-0.032, 0.059]$) stays inside the trained range of Section 2.2, while still producing genuinely large deformation (a tip displacement of 3.1–3.3 m on this 48–60 m panel, depending on which of the seven rows of Table 7 below is asked).

Setup. The panel is meshed, generated directly for this geometry rather than imported, with 128 six-node quadratic triangles (384 Gauss points, $n_x = n_y = 8$), coarser than this project’s usual mesh density, for

tractability, since even at this resolution, nesting the true, non-reduced RVE FOM at every Gauss point (the FOM-FE² row of Table 7) costs on the order of a day of wall-clock time (see below). All seven rows use the identical mesh, load ramp, and Total-Lagrangian Newton-Raphson driver, built on this project’s own Kratos Multiphysics [5] assembler (Section 2.1), differing only in which function supplies the stress and tangent at each Gauss point: each trained PANN’s own automatic-differentiation stress and tangent (Section 6.3, verified against finite differences before use: relative tangent error 1.1×10^{-10} for the ICNN and ICKAN cores, 1.4×10^{-4} for the float32 free and regression models) for the first four rows; the consistent, non-finite-difference analytic tangent derived in Appendix A.4 for D-HPROM-ANN and HPROM-ANN; and a central finite-difference tangent, the only option available, for the true FOM itself. Every load step uses the same convergence tolerance (relative residual 10^{-4}) and the same secant line search used throughout this project’s solvers, except for the true-FOM row, run without line search (plain Newton already converges it, and line search would triple that row’s own already-considerable cost). The *S*-error column below compares each row’s own native homogenization output directly against FOM-FE²’s, both computed the same way, a full volume average for FOM-FE² and its MAW-ECM-sparse approximation (37) for HPROM-ANN and D-HPROM-ANN; this isolates hyperreduction and manifold-regression error cleanly, so the reaction-force correction of Section 6.3 is deliberately not applied to either side here.

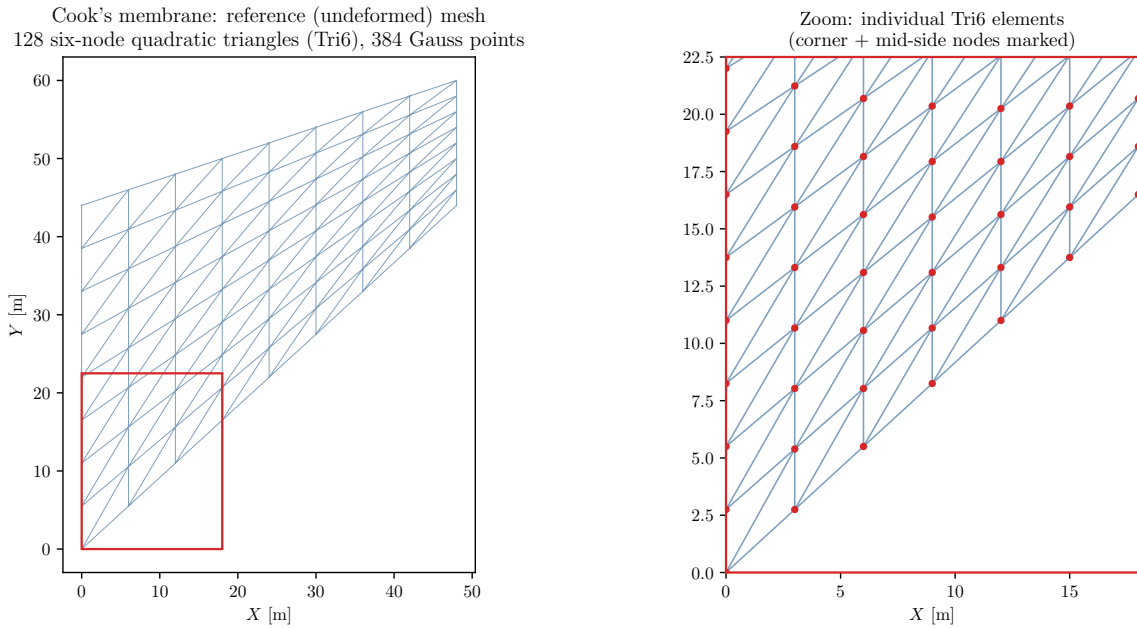


Figure 17: Cook’s membrane, reference (undeformed) mesh, with a zoomed inset showing individual elements directly: each is a six-node quadratic triangle (Tri6), corner and mid-side nodes both marked, two sharing the diagonal of every coarse cell.

Results. The four PANN tiers, D-HPROM-ANN, and HPROM-ANN behave exactly as their test-trajectory accuracy (Table 6) would suggest, and the true FOM, once nested the same way at every Gauss point, joins the same qualitative pattern. The two certified cores, both ROM/HPROM surrogates, and the true FOM itself converge at every one of the 20 load steps, in exactly three Newton iterations each, with no change in iteration count from the smallest to the largest load. The two uncertified tiers both fail to converge already at the *first* load step, only 5% of the final load, but not in the same way, and, importantly, without either one diverging, since neither residual grows without bound or produces a non-finite state, in contrast to the outright divergence As’ad et al. [3] and Klein et al. [1] report for their own uncertified baselines. Instead, each stalls at its own small but non-vanishing plateau; following the same protocol throughout, both are pushed through the entire 20-step ramp regardless, capping every step at 30 iterations and simply carrying whatever iterate resulted into the next, larger load step, and reach a bounded final state without ever diverging. A direct rank-one convexity check at the exact Gauss-point states each law’s own Cook run visits (the energy-free curvature formula of Appendix A.4, since two of the six laws have no energy potential to differentiate twice) finds no violation anywhere, including at the free model’s own stalled state. This failure is a genuine structural consequence of the missing certificate, not one a pointwise check at the solution itself would catch. Table 7 reports the full seven-way comparison: Newton behavior, final tip displacement, and wall-clock time for the complete 20-step ramp on this project’s own workstation (no parallelization attempted), together

with each row’s speedup and accuracy against the true FOM.

Table 7: Cook’s membrane at $n_x = n_y = 8$ (128 elements, 384 Gauss points), full 20-step ramp, every model this paper has trained or built as the macroscopic material law, plus the true, non-reduced FE² solve (first row, FOM-FE²) nested the same way at every Gauss point, as the actual multiscale ground truth this whole hierarchy approximates and the baseline the last three columns measure against. Tip u_y err. is the relative error of the tip-edge range’s midpoint, the single number the range column itself reports as a range; S err. is the relative L^2 error of the full 384-Gauss-point final-state stress field. The two ROM/HPROM rows use the float64, consistent-tangent variant of each surrogate (Appendix A.4).

Model	Newton behavior	Tip u_y range [m]	Wall time	Speedup	Tip u_y err.	S err.
FOM-FE ² [†]	converged, 3 it./step	3.0777–3.2116	76 992.2 s	–	–	–
Regression (tier 1)	stalled, every step [†]	3.070–3.202	0.4 s [‡] [§]	1.9×10^5	0.28%	0.12%
Free hyperelastic (tier 2)	stalled, every step [†]	3.114–3.251	1.3 s [‡] [§]	5.9×10^4	1.20%	2.90%
Polyconvex ICNN (tier 3a)	converged, 3 it./step	3.0979–3.2330	2.6 s [§]	3.0×10^4	0.66%	0.29%
Polyconvex ICKAN (tier 3b)	converged, 3 it./step	3.1678–3.3019	8.7 s [§]	8.9×10^3	2.87%	4.62%
HPROM–ANN-FE ²	converged, 3 it./step	3.0700–3.2019	1 979.8 s	38.9	0.28%	0.37%
D-HPROM–ANN-FE ²	converged, 3 it./step	3.0958–3.2284	386.4 s	199.3	0.55%	0.60%

[†]Pushed through the full ramp regardless of per-step convergence; the short wall time reflects failing fast, not efficient convergence, and is not comparable to the converged rows. [‡]The full, non-reduced 990-element RVE, nested at every one of the 384 Gauss points with no surrogate or reduction of any kind, its tangent obtained by central finite differences (no analytic microscale tangent is available); wall time is the actual, measured total for the complete 20-step ramp, not an extrapolation; being the ground truth itself, it has no speedup or error entries of its own. [§]Mean of 10 independent repetitions (standard deviation 0.08–0.45 s across the four rows), not a single run, for a more representative wall time.

Cook's membrane: real deformed states, every row of Table 7

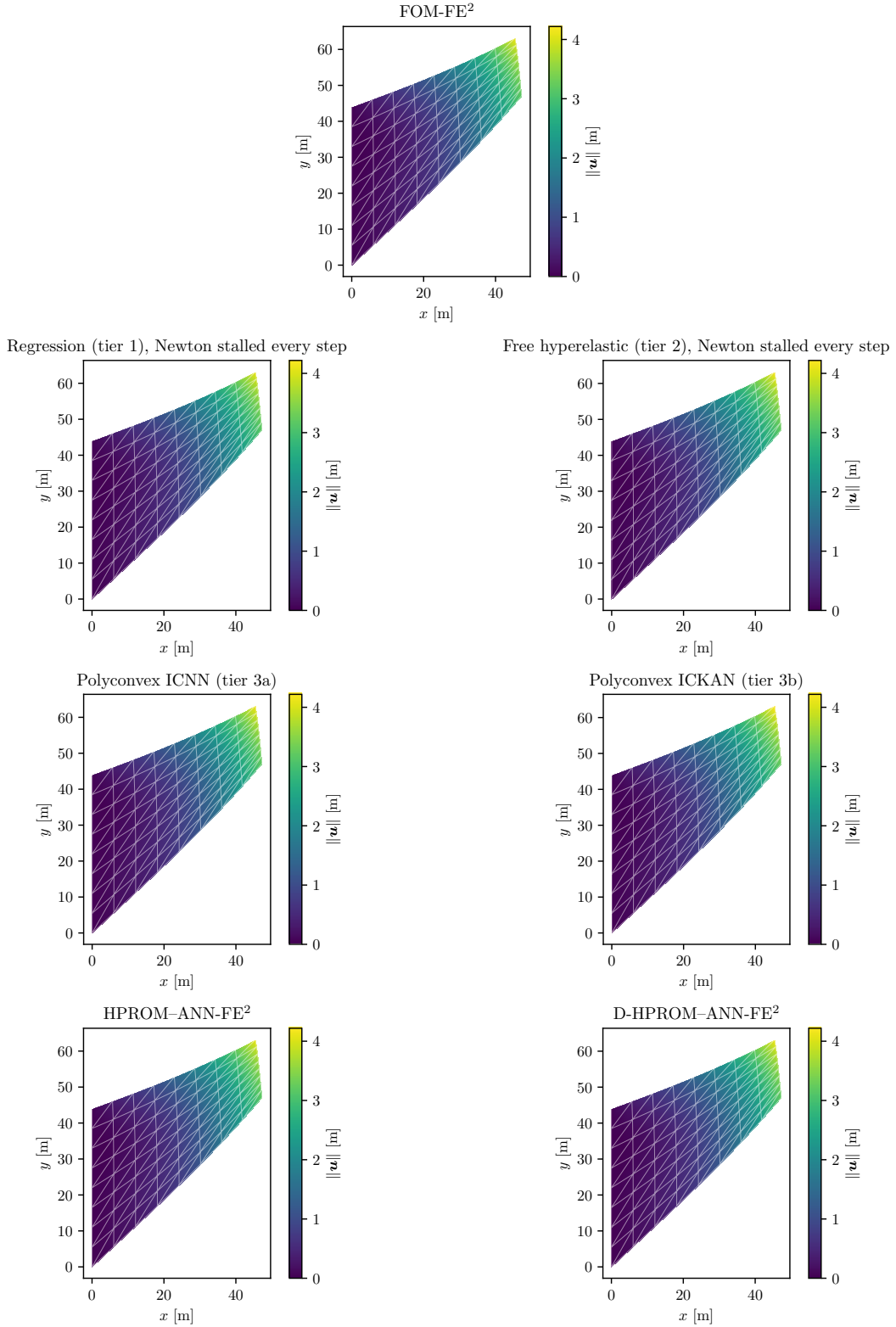


Figure 18: Cook's membrane, real deformed configurations at full load, colored by displacement magnitude, every row of Table 7 (true FOM first, centered). Every row, including the two uncertified tiers – pushed through the full ramp despite never converging per-step, not stopped at their own first stalled step – is visually indistinguishable from the true FOM.

The uncertified tiers' wall time is misleadingly small. It reflects failing fast, twenty stalled load steps with no Newton progress beyond a fixed plateau, not efficient convergence. Among the rows that do converge, the

price of fidelity to the true microscale physics is stark and entirely monotonic with how much of that physics each row actually resolves online: the certified/ICKAN cores, a fully-amortized surrogate for the whole microscale problem, cost seconds; D-HPROM-ANN and HPROM-ANN, each solving, or, for D-HPROM-ANN, directly reconstructing, a genuine reduced state at every Gauss point and every Newton iteration, cost minutes to half an hour; the true, non-reduced FOM, solving the actual 990-element RVE from scratch at every one of those same calls, costs the better part of a day (21.4h). This is the real price of the FOM-FE² ground-truth row of Table 7, and exactly why it is only computed once, here, rather than treated as a routine comparison point throughout this paper. Scaling this kind of nested computation further – a costlier RVE, more scales, or both – would likely need to lean on strategies such as As’ad and Farhat’s [4] staggered training, shown for a three-scale (macroscale-mesoscale-microscale) woven fabric model to keep offline cost scaling linearly rather than exponentially in the number of scales involved; the comparison above does not need it at this project’s own scale.

Speedup and accuracy against the true answer. The same table’s speedup and error columns make the cost/accuracy trade quantitative. Every row, certified or not, matches the true FOM’s tip displacement to within 2.9% and its full 384-point stress field to within 4.7%. The certified cores reach this accuracy roughly four orders of magnitude faster than the true FOM (speedups of $0.9\text{--}3.0 \times 10^4$); both ROM/HPROM surrogates trade most of that speedup, to $39\text{--}199\times$, for the tightest accuracy of any row. That the uncertified tiers are comparably accurate here is not a contradiction of Section 6.5’s own point but a restatement of it, because accuracy alone does not expose a missing certificate. What the missing certificate actually costs, on this problem, is the Newton solve itself failing to converge (the Results paragraph above), not the accuracy of whatever state the solver is eventually pushed to.

Cook's membrane: equivalent (von Mises) stress field, every row of Table 7

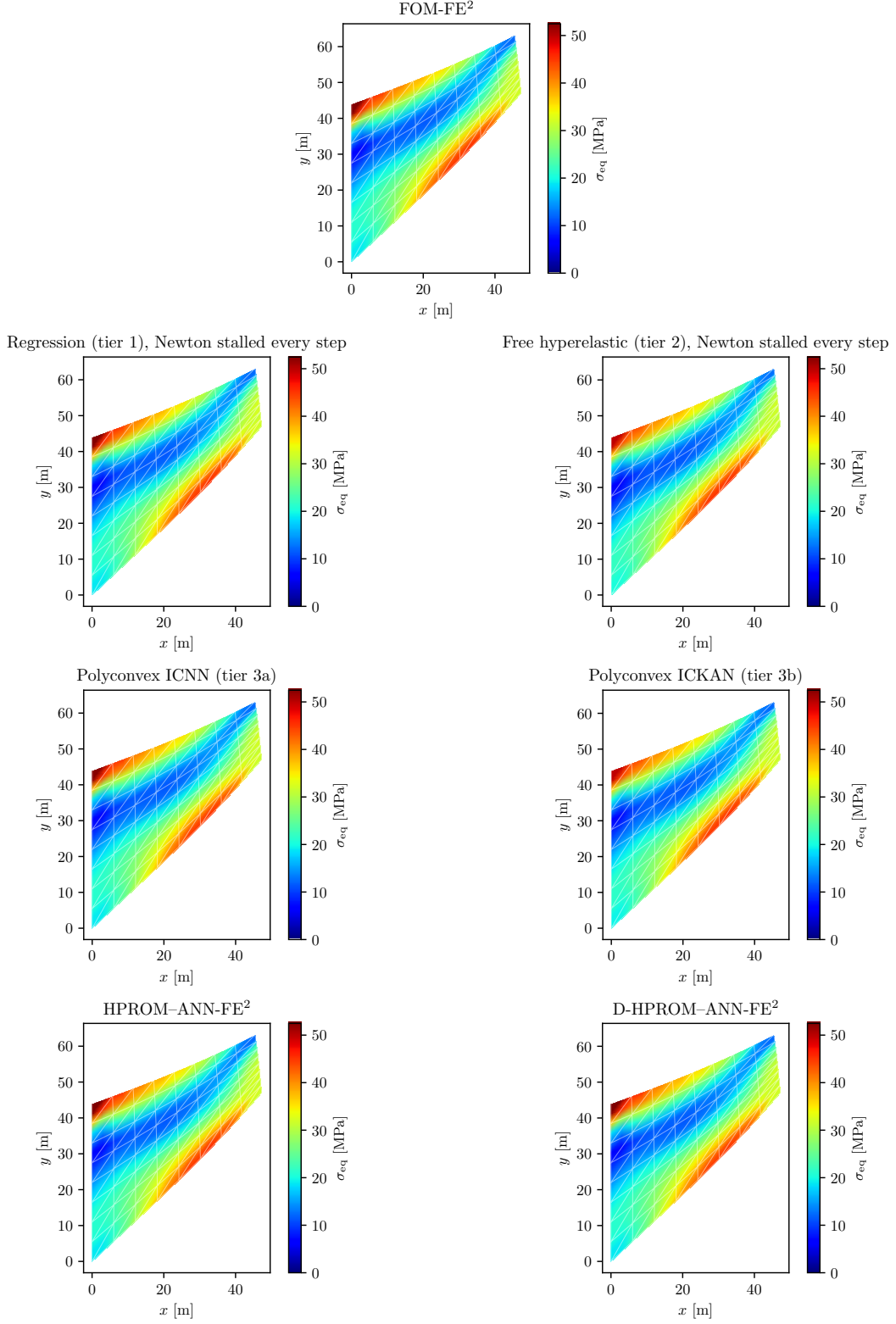


Figure 19: Cook's membrane, equivalent (von Mises) stress field at each row's own reported final state, same rows and layout as Fig. 18. The stress hotspot near the clamped top-left corner, and its overall spatial pattern, match closely across every row, including the two uncertified tiers – consistent with Table 7's own full-field error, under 5% for every row.

Small-strain accuracy and a corrective loss term. The certified cores’ accuracy is not uniform across the strain range. At the small strains Cook’s own first load step visits ($|\varepsilon|$ from 6×10^{-7} to 3×10^{-3}), both carry an 8–15% stress error against the true FOM, an order of magnitude worse than their test-trajectory error (Table 6); the two ROM/HPROM surrogates and the uncertified tier-2 free model all stay within 3% there. We attribute this to two compounding factors. The loss (38) normalizes the stress term by the aggregate mean-squared target *over the whole training set* (Section 6.1), so small-stress, small-strain states carry almost no training signal; and, since the free model does not share this weakness despite an identical loss, the input-convexity constraint itself costs some representational flexibility near the reference state, a real price of the certificate distinct from the aggregate accuracy cost already reported as negligible.

Only the loss-weighting factor is directly correctable, without touching the polyconvexity construction, the features, or the training data. We add one further term,

$$\mathcal{L}_{\text{tan0}} = \frac{\|\mathbb{C}_\theta(\mathbf{0}) - \mathbb{C}_0\|^2}{\|\mathbb{C}_0\|^2}, \quad \mathcal{L} = 0.65 \mathcal{L}_W + \mathcal{L}_{S,\text{agg}} + w_{\text{tan0}} \mathcal{L}_{\text{tan0}}, \quad (42)$$

matching the model’s own tangent at zero strain, $\mathbb{C}_\theta(\mathbf{0})$, against the RVE’s true linear-elastic-limit tangent $\mathbb{C}_0$ – the one reference value in this paper needing no nonlinear FOM solve at all, since three linear RVE solves, one per unit macro-strain direction, give it exactly. Retraining the certified ICNN and ICKAN cores with $w_{\text{tan0}} = 5$ (found by a small sweep, Appendix A.5, not fine-tuned further) drops their first-load-step error to 0.2–1.3% and 0.9–2.8% respectively; this is the version Table 7 reports throughout. The correction has a real cost elsewhere; both models’ error on the full test trajectory of Table 6 rises from sub-percent to the low single digits, since the new term competes for the same limited model capacity. Since that metric is not sensitive to the small-strain regime this term targets, Table 6 keeps the original, $w_{\text{tan0}} = 0$ cores; Table 8 quantifies the trade directly.

Table 8: The small-strain correction’s effect on Cook’s membrane, $n_x = n_y = 8$: tip u_y range for the two certified cores as originally trained ($w_{\text{tan0}} = 0$) versus corrected via (42) at $w_{\text{tan0}} = 5$, the version Table 7 reports. Both converge in exactly three Newton iterations at every load step either way.

Model	Tip u_y range [m], original	Tip u_y range [m], corrected (Table 7)
Polyconvex ICNN (tier 3a)	3.4705–3.6070	3.0979–3.2330
Polyconvex ICKAN (tier 3b)	3.5222–3.6593	3.1678–3.3019

7 Limitations and future work

Three scope limits of the present study are worth stating plainly, since each bears directly on how far the claims above extend.

The construction of Section 4.6.4 assumes no material symmetry group, and the balanced direction systems are not tied to one; this is a property of the construction itself. The specific RVE used to validate it, however, a square unit cell with a centered circular hole (Fig. 2), does have D_4 symmetry, so the test-trajectory accuracy of Section 6.3 does not by itself demonstrate the construction’s behavior on a genuinely asymmetric microstructure. Repeating the same protocol on an RVE with no symmetry at all, an off-center or elliptical hole, or a random inclusion pattern, would close this gap directly.

A second gap is more theoretical. As noted in Section 4.6.4, this paper does not prove that two distinct admissible $\mathbf{C}$ always produce distinct feature vectors (29), nor that three direction systems are the minimum or optimal number. The construction is verified sufficient for this RVE’s own test-trajectory response, not shown to be representationally optimal or complete for a general anisotropic material.

Finally, some ablations and cost figures are missing. Sensitivity to the number of direction systems (one or two instead of three), to the specific angles within a system, and to the random seed used for training are not reported here, nor is the online evaluation cost of the certified PANN relative to a direct microscale solve or to HPROM–ANN, which would be needed to argue for the certified construction on efficiency grounds rather than admissibility grounds alone. The test-trajectory comparison of Section 6.3 also uses a single test trajectory; a broader set of test paths, including states well outside the convex hull of the training trajectories, would give a fuller picture of extrapolation behavior.

None of this weakens the results already reported, because every admissibility proof in Sections 4.2–4.7 holds unconditionally, independent of which RVE, which direction systems, or which downstream solver is used. These are open extensions, and we would rather name them here than leave a reader to find them.

Section 6.6 already takes the further step of nesting D-HPROM–ANN, HPROM–ANN, and the true, non-reduced RVE FOM itself as the RVE-level solver inside the macroscopic Newton loop of its own benchmark,

at every macroscopic Gauss point, rather than deploying any surrogate directly, at one fixed mesh ($n_x = 8$, 384 Gauss points) shared by all seven rows of Table 7. We do not report a mesh-convergence study of this benchmark, and make no claim that its macroscopic answer is itself resolution-independent; that question is orthogonal to the one the comparison is built to ask. Its purpose is to rank fidelity tiers – fully-amortized surrogate, reduced, and true FOM – against one another at a single shared macroscopic discretization, not to pin down the panel’s converged tip displacement in absolute terms, and every row is nested at the same mesh precisely so that the comparison among them stays meaningful regardless of where that mesh sits on its own convergence curve.

8 Conclusions

Every item of the admissibility checklist in Section 3 is addressed with a closed-form argument rather than a numerical penalty. (C1) holds for any energy-derivative stress definition; (C2) holds because the model is built from $\mathbf{C}$ and $\text{cof}(\mathbf{C})$; (C3)’s actual content, tensor symmetry, holds by parameterizing only the three independent stress components, while material symmetry (D_4) is deliberately left unimposed; (C4) holds exactly, by an additive-constant argument for the energy and by the balanced-direction cancellation of Proposition 2 for the stress; (C5), correctly stated as Ball’s (22) rather than the stronger, and for this RVE false, SPD-tangent shorthand, is proved by the explicit convex construction of Proposition 3, with the weaker, correct rank-one necessary condition verified numerically; and (C6) is supplied by the sixth-power directional terms together with the logarithmic/quadratic volumetric barrier. None of (C1)–(C6) depends on the loss weighting of Section 6.1, an orthogonal, purely statistical choice that happened to improve every reported test-trajectory metric, including S_{12} ’s own componentwise error, contrary to the naive expectation that dropping its dedicated weighting should trade it away for a better fit of S_{11} , S_{22} . Any future retraining that changes the relative magnitude of S_{12} against S_{11} , S_{22} , for a different RVE geometry or load protocol, say, should revisit the component-stress weight rather than assume it transfers unchanged.

The construction is also not tied to one convex architecture. Section 6.4 shows the identical proofs, and an equally clean reference-stress cancellation at machine precision, carrying over to an input-convex Kolmogorov–Arnold core, at comparable test-trajectory accuracy to the ICNN core with far fewer parameters.

The remaining two results sections make the case for a certificate, rather than an accuracy check, from two independent directions, using the four-tier ablation of Section 4.2.1: pure regression, free hyperelastic ANN, polyconvex ICNN, polyconvex ICKAN. Section 4.2.1 shows a direct, quantitative violation. The tier-1 pure-regression model has no energy potential at all, and a trapezoidal cyclic-work test $\oint_{\mathcal{L}} \mathbf{S} \cdot d\boldsymbol{\varepsilon}$ around closed strain loops, refined to distinguish a genuine curl from quadrature error, finds a stable, non-vanishing residual for it, and a residual that vanishes under refinement for the three energy-based tiers – a direct, measured failure of (C1). Section 6.5 shows the analogous failure one level up the hierarchy, for (C5). An otherwise identically trained, uncertified flexible PANN (tier 2) matches or slightly beats the certified models on every reported accuracy metric, while failing the rank-one necessary condition of polyconvexity on 7.400% of sampled states, at ordinary, non-extreme deformations. In both cases the accuracy metrics of Table 6 are silent, or actively misleading, since the tier-1 pure-regression model is in fact the most accurate of the four ANNs on the test trajectory, by every reported metric. Accuracy and the admissibility certificates simply answer different questions. Only a direct, targeted test of each checklist item can confirm the certificate is actually present.

The same distinction carries over to an actual structural solve. Deployed as the macroscopic material law of a Cook’s-membrane benchmark (Section 6.6), the certified ICNN and ICKAN cores converge in three Newton iterations at every one of twenty ramped load steps, up to a genuinely large tip deflection; the free model, accurate on the same test trajectory metrics as the certified cores, fails to converge at the very first, smallest step – a real structural consequence of the missing certificate, even though a rank-one convexity check at the exact Gauss-point states it visits finds no violation there, only in the aggregate global sample of Section 6.5. The certificate’s practical value is not confined to the standalone constitutive surrogate.

Nesting the projection-based ROM/HPROM surrogates of Section 5, and the true, non-reduced RVE FOM itself, as the actual RVE-level solver at every macroscopic Gauss point – a genuine FE^2 computation, not a direct deployment – confirms this is not an artifact of deploying the certified PANN directly rather than solving a genuine microscale problem. Both ROM/HPROM surrogates converge in the same three Newton iterations as the certified cores, once matched to the arithmetic precision the small-strain regime requires, and the same energy-free rank-one test, extended to all six model variants without needing a potential, finds the same architectural split, zero violations for the two polyconvex-by-construction cores and a real double-digit rate for every model without that guarantee, sampled broadly, even though none of the six shows a violation at the particular states this specific structural problem visits. Comparing every model directly against the true FE^2 answer this way, rather than only against each other, further exposes a small-strain accuracy gap

specific to the certified cores, invisible to their own test-trajectory metric and traced to how the training loss weights small-stress states; one additional, cheap loss term, matching each model’s own tangent stiffness at zero strain to the RVE’s true linear-elastic limit, closes most of that gap without touching the admissibility construction itself.

A Supplementary derivations

This appendix collects material that is honest and, in places, important, but not needed to follow the main argument: an algebraic aside on the feature set, a closer look at one competing paper’s own internal logical structure, and the periodic-boundary-condition averaging identity in full. Nothing here is used by any proof in the main text.

A.1 Algebraic redundancy of the Q^H features

This expands Remark 2. Writing $T = \text{tr } \mathbf{C}$ and $t_{ki} = \mathbf{d}_{ki} \cdot \mathbf{C} \mathbf{d}_{ki}$, the elementary 2×2 identity $\text{cof}(\mathbf{C}) = T\mathbf{I} - \mathbf{C}$ (the same identity behind Lemma 1, applied to $\mathbf{C}$ itself rather than to a rank-one $\mathbf{d} \otimes \mathbf{d}$) gives $\mathbf{d}_{ki} \cdot \text{cof}(\mathbf{C}) \mathbf{d}_{ki} = T - t_{ki}$, and hence, using $\sum_i w_{ki} = 2$ and $\sum_i w_{ki} t_{ki} = T$ (both consequences of the balance condition (26) established in the proof of Proposition 1),

$$Q_{k,2}^H = Q_{k,2}^F, \quad Q_{k,3}^H = -T^3 + 3T Q_{k,2}^F - Q_{k,3}^F, \quad (43)$$

exactly, for every system k and every $\mathbf{C}$ (verified symbolically and confirmed numerically to machine precision for random balanced systems and random $\mathbf{C}$). So the fifteen entries of $\mathbf{z}_{\text{pc}}$ in (29) are not fifteen algebraically independent scalars: given $I_1 = T$ and $\{Q_{k,2}^F, Q_{k,3}^F\}_{k=1}^3$ (seven numbers), the corresponding $\{Q_{k,2}^H, Q_{k,3}^H\}_{k=1}^3$ are fixed exactly by (43).

The redundancy does not, however, make $Q_{k,3}^H$ useless as a separate network input, for an architectural reason specific to the monotone ICNN of Section 4.6.4: reconstructing $Q_{k,3}^H$ from $T, Q_{k,2}^F, Q_{k,3}^F$ via (43) requires the *negative* coefficients -1 on T^3 and -1 on $Q_{k,3}^F$, which a network constrained to non-negative weights at every layer (needed for the convexity-composition argument of Proposition 3) cannot represent internally. Supplying $Q_{k,3}^H$ directly as its own coordinate therefore gives the monotone network access to a combination of $\mathbf{C}$ that no non-negative combination of the other coordinates could reconstruct, even though it carries no additional information in the unconstrained, purely algebraic sense: much as a monotone function of x and, separately, of $-x$ is strictly more expressive than a monotone function of x alone.

$Q_{k,2}^H$ is a different case, and we state this plainly rather than let the argument above appear to cover it as well. The first identity in (43) has coefficient $+1$, with no negative sign to obstruct reconstruction. A monotone network already has direct, unconstrained access to $+Q_{k,2}^F$ with an arbitrary non-negative coefficient, so assigning independent non-negative weights to $Q_{k,2}^F$ and to $Q_{k,2}^H = Q_{k,2}^F$ is exactly equivalent to assigning their sum to $Q_{k,2}^F$ alone. Supplying $Q_{k,2}^H$ as a separate coordinate therefore changes nothing about the class of functions the trained network can represent; unlike $Q_{k,3}^H$, it is not merely redundant in information but redundant in *expressiveness* too. It is retained in the deployed architecture for uniformity with the other per-system coordinates, not for any representational benefit, and we report this plainly rather than imply the same architectural justification covers both cases.

A.2 A closer reading of Klein et al.’s trivial-group case

This expands the pointer in Section 4.6.4. Klein et al.’s [24] own Table 1 lists the *trivial* group C_1 ($|\mathcal{G}| = 1$, identity only) among the finite groups their construction covers, and substituting $\mathcal{G} = C_1$ into their group-symmetrization operator collapses the symmetrization sum to its single, un-symmetrized term. Read this way, their own machinery already contains the seed of a fully general, no-symmetry construction satisfying (C1), (C2), (C4)–(C6), a nice illustration of just how general their underlying integrity basis is. They do not state, highlight, verify, or numerically exercise this reading anywhere in their paper, since every one of their worked examples (their Section 5) uses the cubic group specifically. Crediting them with having solved the general case would therefore over-credit an inference they never made or checked, which is why the main text states our own contribution as complementary rather than as closing a gap no other construction could possibly reach.

A.3 The periodic-boundary-condition averaging identity, in full

This expands the pointer in Section 2.1. Under periodic boundary conditions, a classical boundary-integral identity (the divergence theorem applied to the periodic fluctuation field) gives $\langle \mathbf{F} \rangle = \bar{\mathbf{F}}$ exactly, where $\bar{\mathbf{F}}$

is the applied macroscopic deformation gradient and $\langle \cdot \rangle$ denotes the volume average over the RVE, and the same argument applied to the microscopic first Piola–Kirchhoff stress gives a volume-averaged macro-stress that is automatically energy-conjugate. Because the Green–Lagrange strain $\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$ is a *nonlinear*, quadratic function of $\mathbf{F}$, this identity does not by itself give $\langle \mathbf{E} \rangle = \mathbf{E}(\bar{\mathbf{F}})$; the two differ, in general, by a Jensen-type gap driven by the fluctuation $\mathbf{F} - \bar{\mathbf{F}}$. The exact, elementary averaging identity that transfers without approximation under periodicity is the one stated above, at the level of $\mathbf{F}$, not of $\mathbf{E}$.

A.4 A consistent tangent for the ROM/HPROM surrogates, and an energy-free rank-one test

This expands the brief description in Section 6.6. Neither operating mode of Section 5’s hyper-reduced pipeline is used inside a Newton solve anywhere else in this paper (Section 6.3 only evaluates both along a fixed, already-known test trajectory), so no tangent for either one is derived there. Nesting them as the macroscopic material law of Section 6.6 requires one: both the macroscopic Newton–Raphson iteration and the line search it uses need $d\mathbf{S}_{\text{macro}}/d\boldsymbol{\varepsilon}$, not only $\mathbf{S}_{\text{macro}}(\boldsymbol{\varepsilon})$, at every Gauss point and every macro-iteration.

Write the reconstruction map shared by both modes, from (36), as $\mathbf{u}_f(\mathbf{q}; \boldsymbol{\varepsilon}) = \mathbf{u}_f^{\text{aff}}(\boldsymbol{\varepsilon}) + \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathcal{N}(\mathbf{q})$, and let

$$\mathbf{J}(\mathbf{q}) := \frac{\partial}{\partial \mathbf{q}}(\mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathcal{N}(\mathbf{q})) = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}}{\partial \mathbf{q}} \in \mathbb{R}^{n_L \times n} \quad (44)$$

be its own tangent, the curved-manifold analogue of the fixed $\mathbf{V}_{\text{tot}}$ used by the linear PROM; $\partial \mathcal{N} / \partial \mathbf{q}$ is obtained once per query by automatic differentiation of the trained decoder, the same tool already used for every constitutive tangent in Section 6.3. D-HPROM–ANN sets $\mathbf{q} = \boldsymbol{\mu}(\boldsymbol{\varepsilon}) := \mathbf{A}\boldsymbol{\varepsilon} + \mathbf{c}$, the fixed, offline-fitted affine map of Section 5.1 (the same map used to initialize HPROM–ANN’s own Newton correction); HPROM–ANN instead solves the ECM-weighted, Galerkin-projected residual of (37), restricted to the primary coordinate,

$$\mathbf{r}_r(\mathbf{q}; \boldsymbol{\varepsilon}) := \sum_{g \in \mathcal{Z}} \mathbf{J}(\mathbf{q})^T \mathbf{B}_{e,g}^T \mathbf{S}_g(\mathbf{u}_f(\mathbf{q}; \boldsymbol{\varepsilon})) \omega_g = \mathbf{0}, \quad (45)$$

for $\mathbf{q}^*(\boldsymbol{\varepsilon})$ by Newton iteration, starting from $\boldsymbol{\mu}(\boldsymbol{\varepsilon})$.

D-HPROM–ANN: direct chain rule. Since $\mathbf{q} = \boldsymbol{\mu}(\boldsymbol{\varepsilon})$ is explicit, $d\mathbf{u}_f/d\boldsymbol{\varepsilon} = \partial \mathbf{u}_f^{\text{aff}} / \partial \boldsymbol{\varepsilon} + \mathbf{J}(\boldsymbol{\mu}(\boldsymbol{\varepsilon}))\mathbf{A}$ directly, with $\mathbf{A} := d\boldsymbol{\mu}/d\boldsymbol{\varepsilon}$ the fixed offline coefficient matrix. The macroscopic output itself is one further, adaptively-weighted average, $\mathbf{h}(\boldsymbol{\varepsilon}) = |\Omega_0|^{-1} \sum_e A_e \omega_{\sigma,e}(\boldsymbol{\varepsilon}) \bar{\mathbf{v}}_e(\boldsymbol{\varepsilon})$ (either $\mathbf{h} = \mathbf{S}_{\text{macro}}$ or the homogenized strain, both of this same form, and both restricted in practice to $\mathcal{Z}_\sigma$), so a second product rule applies,

$$\frac{d\mathbf{h}}{d\boldsymbol{\varepsilon}} = \frac{1}{|\Omega_0|} \sum_e A_e \left[\frac{d\omega_{\sigma,e}}{d\boldsymbol{\varepsilon}} \bar{\mathbf{v}}_e + \omega_{\sigma,e} \frac{d\bar{\mathbf{v}}_e}{d\boldsymbol{\varepsilon}} \right], \quad (46)$$

where $d\bar{\mathbf{v}}_e/d\boldsymbol{\varepsilon}$ is element e ’s own Gauss-point mean of $\mathbb{C}_g \mathbf{B}_{e,g}(d\mathbf{u}_f/d\boldsymbol{\varepsilon})_e$ ($\mathbb{C}_g := \partial \mathbf{S}_g / \partial \mathbf{E}_g$, the true microscale Neo-Hookean tangent, already assembled at every Gauss point for the stress itself), and $d\omega_{\sigma,e}/d\boldsymbol{\varepsilon}$ is the trained MAW–ECM weight regressor’s own Jacobian (Section 5.2), again by automatic differentiation. Every factor above is closed-form or autograd-exact; none is finite-differenced.

HPROM–ANN: the implicit step. Now $\mathbf{q}^*(\boldsymbol{\varepsilon})$ solves (45) identically in $\boldsymbol{\varepsilon}$; differentiating that identity and applying the implicit function theorem,

$$\frac{d\mathbf{q}^*}{d\boldsymbol{\varepsilon}} = - \left(\frac{\partial \mathbf{r}_r}{\partial \mathbf{q}} \right)^{-1} \frac{\partial \mathbf{r}_r}{\partial \boldsymbol{\varepsilon}} =: \mathbf{K}_r^{-1} \frac{\partial \mathbf{r}_r}{\partial \boldsymbol{\varepsilon}}, \quad \mathbf{K}_r := - \frac{\partial \mathbf{r}_r}{\partial \mathbf{q}} \Big|_{\mathbf{q}^*}, \quad (47)$$

where $\mathbf{K}_r$ is exactly the matrix the Newton corrector already assembles for its own last update, $\Delta \mathbf{q} = \mathbf{K}_r^{-1} \mathbf{r}_r$: it already includes the manifold’s own curvature (a second-derivative correction to (44), since $\mathbf{J}(\mathbf{q})$ is not constant, absent from a Gauss–Newton approximation but present in a full Newton step) and the residual weights’ own dependence on $\mathbf{q}$, so nothing further needs to be assembled for it. The right-hand side is comparatively simple: the decoder and the residual weights ω_g depend on $\boldsymbol{\varepsilon}$ only through $\mathbf{q}$, so $\boldsymbol{\varepsilon}$ enters $\mathbf{r}_r$ directly only through the affine lifting $\mathbf{u}_f^{\text{aff}}(\boldsymbol{\varepsilon})$,

$$\frac{\partial \mathbf{r}_r}{\partial \boldsymbol{\varepsilon}} \Big|_{\mathbf{q}} = - \sum_{g \in \mathcal{Z}} \omega_g \mathbf{J}(\mathbf{q})^T \mathbf{B}_{e,g}^T \mathbb{C}_g \mathbf{B}_{e,g} \frac{\partial \mathbf{u}_f^{\text{aff}}}{\partial \boldsymbol{\varepsilon}}, \quad (48)$$

the same already-assembled element tangent contracted with the affine lifting’s own, closed-form strain sensitivity ($\partial \mathbf{F} / \partial \boldsymbol{\varepsilon}$). Once $d\mathbf{q}^*/d\boldsymbol{\varepsilon}$ is known from (47), the outer chain rule (46) applies unchanged, with

$\mathbf{J}(\mathbf{q}^*) d\mathbf{q}^*/d\boldsymbol{\varepsilon}$ in place of $\mathbf{J}(\boldsymbol{\mu}(\boldsymbol{\varepsilon}))\mathbf{A}$: the two operating modes share their entire outer, homogenization-level derivative, differing only in how $d\mathbf{q}/d\boldsymbol{\varepsilon}$ is obtained.

Both tangents were checked against central finite differences on the real trained models (not a synthetic sanity check) before use: relative error 6×10^{-4} – 8×10^{-4} for D-HPROM-ANN and 4×10^{-5} – 2×10^{-3} for HPROM-ANN, at strains at or below this project’s own training range; D-HPROM-ANN’s degrades to a few percent at strains far outside it ($|\boldsymbol{\varepsilon}| \sim 1$, roughly twenty times any strain Section 6.6’s benchmark actually develops), consistent with reduced trust in the trained decoder’s own derivative that far into extrapolation rather than an error in (47)–(48) itself.

An energy-free rank-one test. Section 4.6.3’s rank-one check is stated as an energy second derivative, $\frac{d^2}{dt^2} W(\mathbf{F}_0 + t \mathbf{a} \otimes \mathbf{b})|_{t=0} \geq 0$, which presupposes an energy potential W : unavailable for the tier-1 regression model (Section 4.2.1) and for both ROM/HPROM surrogates, which reconstruct stress directly. Write $\mathbf{A} := \mathbf{a} \otimes \mathbf{b}$ and $\mathbf{F}(t) = \mathbf{F}_0 + t\mathbf{A}$; since $\mathbf{C}(t) = \mathbf{F}(t)^T \mathbf{F}(t) = \mathbf{C}_0 + t(\mathbf{F}_0^T \mathbf{A} + \mathbf{A}^T \mathbf{F}_0) + t^2 \mathbf{A}^T \mathbf{A}$ is exactly quadratic in t (no higher terms), so is $\mathbf{E}(t) = \frac{1}{2}(\mathbf{C}(t) - \mathbf{I})$, and its first two t -derivatives at $t = 0$ are closed-form and themselves t -independent,

$$\dot{\mathbf{E}} := \frac{1}{2}(\mathbf{F}_0^T \mathbf{A} + \mathbf{A}^T \mathbf{F}_0), \quad \ddot{\mathbf{E}} := \mathbf{A}^T \mathbf{A}, \quad (49)$$

using $|\mathbf{a}| = 1$ for the second. Define $\varphi(t) := \int_0^t \mathbf{S}(\mathbf{E}(t')) \cdot \dot{\mathbf{E}}(t') dt'$, the work done along this one-parameter path, well-defined whether or not $\mathbf{S}$ derives from a global potential; when it does, $\varphi \equiv W(\mathbf{F}(t)) - W(\mathbf{F}_0)$ exactly, so $\varphi''(0) \geq 0$ is the same condition. Differentiating φ twice via the chain rule, using $d\mathbf{S}/dt = \mathbb{C} \dot{\mathbf{E}}$ at $t = 0$ and (49),

$$\varphi''(0) = \dot{\mathbf{E}} \cdot \mathbb{C}_0 \cdot \dot{\mathbf{E}} + \mathbf{S}_0 \cdot \ddot{\mathbf{E}}, \quad (50)$$

in Voigt notation throughout ($\dot{\mathbf{E}}, \ddot{\mathbf{E}}$ with the usual engineering doubled-shear convention, $\mathbf{S}_0$ undoubled, so every product above is a plain dot product), needing only $\mathbf{S}_0$ and $\mathbb{C}_0 := \partial \mathbf{S} / \partial \mathbf{E}|_0$ at the base state, exactly the two quantities every law in this paper already returns. Validated against the energy-based, double finite-difference method Section 4.6.3 itself uses, for the two certified cores that have both a potential and a tangent to cross-check against, (50) agrees to a relative error at machine precision (10^{-15} – 10^{-16}), before being applied, in Section 6.6, to the three model variants no energy-based check can reach at all.

A.5 The small-strain tangent-term weight sweep

This expands the brief mention in Section 6.6 of how $w_{\text{tan}0} = 5$ in (42) was chosen. Both certified cores were retrained at six weights, $w_{\text{tan}0} \in \{0, 0.3, 1, 2, 3, 5\}$, identical in every other respect to Section 6’s own protocol ($w_{\text{tan}0} = 0$ reproduces the original models of Table 6 exactly); this was a small, one-dimensional sweep run once and not fine-tuned further, not a search over a larger hyperparameter space. Table 9 reports, for every weight tested, the tangent-at-zero relative error $\|\mathbb{C}_\theta(\mathbf{0}) - \mathbb{C}_0\| / \|\mathbb{C}_0\|$ that (42) directly penalizes; the test-trajectory energy and stress relative errors W and S of Table 6; the mean relative stress error, against the true FOM, at Cook’s own actual first-load-step Gauss-point strains (Section 6.6); and the resulting tip-displacement range from a full 20-step Cook run with that checkpoint deployed as the macroscopic law, exactly as in Table 7. Every row converges in three Newton iterations per load step, same as every other certified-core row reported in this paper.

Both cores improve monotonically in the two metrics the new term targets, tangent-at-zero error and the Cook step-1 comparison, as $w_{\text{tan}0}$ increases, and monotonically worsen in test-trajectory S , the expected trade against the same limited model capacity; test trajectory W improves initially and then plateaus (ICNN) or is non-monotonic within a narrow band (ICKAN), consistent with W being only loosely coupled to either competing objective. $w_{\text{tan}0} = 5$, the largest weight tested, gives the best tangent-at-zero and Cook step-1 numbers for both cores; we did not test beyond it; the ICKAN core’s own numbers are uniformly noisier across the sweep than the ICNN core’s, consistent with Section 6.6’s own observation that the ICKAN core’s error is generally somewhat higher and less smooth than the ICNN core’s throughout this paper.

Table 9: The small-strain tangent-term weight sweep, both certified cores. $\mathcal{L}_{\text{tan0}}$ err. is the tangent-at-zero relative error (42); W and S are the test-trajectory relative L^2 errors of Table 6; Cook step 1 is the mean relative stress error against the true FOM at Cook’s own first-load-step Gauss-point strains (Section 6.6); Cook full u_y is the tip displacement range from a full 20-step Cook run with that checkpoint as the macroscopic law. $w_{\text{tan0}} = 0$ reproduces Table 6’s original models exactly; $w_{\text{tan0}} = 5$ is the value used for the corrected cores of Table 8.

Model	w_{tan0}	$\mathcal{L}_{\text{tan0}}$ err.	W	S	Cook step 1	Cook full u_y [m]
Polyconvex ICNN	0	9.72%	0.42%	0.55%	12.61%	3.4705–3.6070
Polyconvex ICNN	0.3	2.02%	1.18%	1.34%	3.14%	3.1880–3.3250
Polyconvex ICNN	1	1.25%	1.07%	1.43%	1.79%	3.1378–3.2741
Polyconvex ICNN	2	0.92%	0.91%	1.51%	1.19%	3.1197–3.2553
Polyconvex ICNN	3	0.75%	0.88%	1.65%	0.89%	3.1082–3.2437
Polyconvex ICNN	5	0.57%	0.88%	1.82%	0.60%	3.0979–3.2330
Polyconvex ICKAN	0	7.67%	0.37%	0.55%	11.79%	3.5222–3.6593
Polyconvex ICKAN	0.3	5.36%	1.40%	2.51%	9.01%	3.4372–3.5744
Polyconvex ICKAN	1	3.44%	2.15%	4.10%	5.60%	3.2801–3.4160
Polyconvex ICKAN	2	2.14%	2.81%	5.35%	3.34%	3.1915–3.3265
Polyconvex ICKAN	3	1.62%	3.00%	5.84%	2.42%	3.1545–3.2891
Polyconvex ICKAN	5	1.28%	1.52%	4.75%	1.73%	3.1678–3.3019

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